

Equilibrium Uniqueness in Entry Games with Combinatorial Actions and Endogenous Pricing^{*}

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Abstract

Combinatorial entry games combine subset choice, within-firm complementarities, and strategic substitutes, but existing uniqueness results either treat entry as binary, assume constant markups, or take the active set as given. We develop a uniqueness framework for combinatorial entry with endogenous Atkeson–Burstein markups. The key object is a decomposition of the threshold-map Jacobian into density dampening, marginal-value sensitivity, and threshold location, which implies a bathtub-shaped contraction modulus in private-cost noise. When its peak is below one, the Bayesian Nash equilibrium is unique for all noise levels, and supermodularity of the single-firm problem is preserved so existing CDCP solvers remain applicable. In a quantitative application to global cloud infrastructure entry (five providers, 42 countries), the calibrated model matches 99.5% of in-sample entry decisions, and endogenous markups sustain 15 additional firm-market pairs and lower consumer surplus by 10.6% relative to a constant-markup benchmark.

Keywords: entry games, equilibrium uniqueness, combinatorial discrete choice, Bayesian Nash equilibrium, variable markups, supermodular games

JEL Codes: C72, C62, C63, L13, F12

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1. INTRODUCTION

Equilibrium multiplicity is a central obstacle in entry games. When entry decisions are strategic substitutes, multiple configurations of active firms can be mutually consistent, creating well-known difficulties for estimation and counterfactual analysis. Three literatures have made progress on different facets of this problem: [Espin-Sanchez et al. \(2023\)](#) prove uniqueness for *binary* entry games with private costs, but cannot handle within-firm complementarities across actions; the combinatorial discrete choice (CDCP) literature ([Jia, 2008](#); [Arkolakis et al., 2023](#)) handles subset selection with complementarities, but assumes constant markups; and the variable-markup literature ([Atkeson and Burstein, 2008](#); [Edmond et al., 2015](#)) endogenizes pricing, but takes the set of active firms as given. In many important settings—multinational production, retail chain location, platform entry—all three features interact simultaneously: firms choose *which subset* of markets to enter, entry changes market structure, and market structure feeds back into markups and entry incentives. Neither the entry game, the CDCP, nor the pricing game can be solved in isolation.

This paper provides a general uniqueness framework for combinatorial entry games with endogenous pricing. The core methodological contribution is a decomposition of the equilibrium-feedback Jacobian into three economically interpretable forces—*density dampening*, *marginal-value sensitivity*, and *threshold location*—that reveals why the activation–share–markup circularity is self-limiting rather than explosive. The resulting contraction modulus has a distinctive “bathtub” shape: private-cost noise disciplines multiplicity at both extremes (large noise smooths responses directly; small noise pushes thresholds into the tails of the cost distribution), so when the peak modulus is below one the Bayesian Nash equilibrium is unique for *all* noise levels. This criterion is verifiable from model primitives without solving the game, providing a modular route to uniqueness in any combinatorial game with a contractive pricing or payoff block.

We illustrate the framework with a quantitative application to global cloud infrastructure entry, where five hyperscaler providers choose which of 42 countries to establish cloud regions. The calibrated model matches 99.5% of entry decisions in sample with a unique equilibrium across all specifications and counterfactuals. Endogenous markups sustain 15 additional firm-market pairs relative to constant-markup pricing and raise ag-

gregate profit by 30%; the welfare decomposition shows the flipside, that AB pricing lowers aggregate consumer surplus by 10.6% relative to a CES benchmark, with a 2.7% net positive total-surplus effect. The game-theoretic results apply to any combinatorial entry game with contractive pricing—retail chains (Jia, 2008), platforms, or technology adoption—and the cloud application demonstrates their quantitative relevance.

The paper makes three main contributions, supported by a quantitative application to hyperscaler cloud infrastructure deployment:

Core contribution: a Jacobian decomposition for uniqueness. On the intensive margin, we derive a spectral-radius condition for pricing contraction based on the interaction matrix of the markup game (Proposition 11). On the extensive margin, when firms have private information on action-specific fixed costs, the combinatorial activation decision becomes a continuous threshold problem whose Jacobian decomposes into three interpretable forces (Proposition 15). The bathtub-shaped contraction modulus yields a global uniqueness criterion: if the complete-information equilibrium is unique and the peak modulus is below one, the BNE is unique for all noise levels (Theorem 18). Two verification strategies make the criterion operational: because the interaction matrix is block-diagonal across markets, the pricing-contraction hypothesis admits a *direct enumeration certificate* on the full activation lattice (Proposition 21), which we apply to the calibrated cloud economy; and because conditioning on a common across-firm cost shock ζ maps the correlated-shock game to the baseline IID game at shifted fixed costs, the same criterion extends state-by-state under moderate shock dependence (Proposition 14).

Supporting result: supermodularity under endogenous markups. We extend Antràs et al.’s (2017) constant-markup supermodularity result to endogenous AB variable markups (Proposition 6). This preserves the lattice-theoretic structure required for CDCP algorithms and for the threshold system’s monotone comparative statics.

Quantitative illustration: cloud infrastructure entry. We calibrate the framework to global cloud infrastructure, following the identification structure of Arkolakis et al. (2023)—fixing demand and firm scale externally, calibrating fixed costs from entry moments—though with a thinner moment basis. The application shows the mechanism has quantitative bite: endogenous markups materially sustain fringe-firm entry, EU compliance-cost

shocks are regressive in market-structure terms, and the binding constraint in digital fragmentation is market access, not trade costs. Appendix F provides synthetic stress tests across the parameter space, and Appendix E compares four algorithm variants with up to 2^{40} candidate subsets per player.

Entry games and equilibrium selection. This paper contributes primarily to the literature on equilibrium uniqueness in entry games. Beginning with [Bresnahan and Reiss \(1991\)](#) and [Berry \(1992\)](#), the empirical entry literature has confronted multiplicity as a barrier to estimation: [Berry](#) restricts equilibrium selection to obtain a tractable likelihood; [Ciliberto and Tamer \(2009\)](#) use partial identification to bound parameters without selecting among equilibria; [Seim \(2006\)](#) introduces endogenous product-type choices, a structure related to combinatorial entry; [Igami and Yang \(2016\)](#) estimate a dynamic entry model with combinatorial product portfolios using simulation methods that must navigate the multiplicity problem. [Espin-Sanchez et al. \(2023\)](#) prove BNE uniqueness in static entry games where each firm makes a *binary* enter/not-enter decision with scalar private types; their threshold system is over $\mathbb{R}^N$ (one cutoff per firm). When each firm instead chooses a *subset* of actions—e.g., which countries to produce in—the binary framework cannot capture within-firm complementarities, and the threshold system grows to $\mathbb{R}^M$ (one cutoff per player-action pair). We extend their result to these combinatorial action spaces, where within-firm actions interact through the CDCP’s supermodularity and payoffs depend on an endogenous pricing equilibrium. The two results are complementary: [Espin-Sanchez et al.](#) cover general payoff functions with binary actions, while we handle combinatorial action spaces under CES/AB payoffs; neither nests the other as a special case. Three results sharpen the distinction. First, a 2×2 worked example (Section 3.4) constructs a configuration where the binary-action threshold system of [Espin-Sanchez et al.](#) cannot be applied, but the combinatorial CDCP structure resolves uniqueness by internalizing within-firm complementarities. Second, the pricing feedback reaches $\rho(\Lambda) = 0.574$ at stressed parameterizations (Appendix F), so the endogenous-markup channel is quantitatively significant. Third, endogenous AB markups *expand* the uniqueness region relative to constant CES markups (Appendix F): markup compression dampens the dominant-firm responses that drive multiplicity under constant markups.

A key negative result motivates the present approach: [Doraszelski and Satterthwaite](#)

(2010) introduce private scrap values in dynamic Ericson–Pakes models, guaranteeing cutoff equilibria but explicitly demonstrating that multiplicity persists—private information alone does not resolve the number of equilibria. In the present setting, the pricing contraction (Proposition 11) provides the additional structure—bounding how activation changes feed back through markups—that converts the threshold system into a contraction. An alternative selection mechanism is the global-games approach of Harrison and Jara-Moroni (2021), who extend iterated dominance arguments to binary-action games with strategic substitutes: dominance regions propagate from extreme types inward until a unique strategy profile survives. Their approach requires binary action spaces and strict payoff asymmetry across player pairs, while the present paper handles combinatorial actions and operates through a different mechanism (pricing contraction rather than dominance propagation).

Quantitative entry with combinatorial or multi-location choice. A parallel quantitative literature estimates entry models where firms choose portfolios of products or locations, typically restricting attention to equilibria that can be computed without resolving multiplicity. Holmes (2011) estimates a dynamic model of Wal-Mart’s spatial diffusion in which store-level economies of density generate combinatorial interdependence across locations, imposing a perfect-foresight restriction to sidestep multiplicity. Nishida (2015) estimates a static multi-store network-choice model for convenience stores in Okinawa, accommodating within-firm density economies and across-firm substitution but selecting among equilibria via a maximum-profit rule. Wollmann (2018) and Fan and Yang (2020) study equilibrium product characteristics and product-line choice in commercial trucks and smartphones, respectively, where firms choose subsets of products from a menu under oligopolistic competition; both rely on equilibrium-selection assumptions to make counterfactuals tractable. Weintraub et al. (2008) propose oblivious equilibrium for many-firm dynamic industries, approximating Markov-perfect behavior by treating the competitive environment as a long-run average—a different route around multiplicity that trades equilibrium exactness for computational scalability. Relative to these, we provide primitive conditions on preferences and cost heterogeneity under which the static combinatorial entry game has a unique BNE, so that counterfactuals in the same class of applications (multi-site/product entry with endogenous pricing) can be computed without invoking

selection rules or many-firm approximations.

Supermodular games. The lattice-theoretic approach connects to the literature on supermodular games (Milgrom and Roberts 1990; Vives 1990; Milgrom and Shannon 1994; Topkis 1998; Amir 1996). The key challenge is that the cross-player interaction exhibits strategic *substitutes*, not complements: when one firm activates more sites, rivals’ activation incentives weaken. We handle this through the squared best-response map, whose monotonicity restores the Tarski fixed-point structure. Athey (2001) establishes existence of pure-strategy equilibria in games whose payoffs satisfy a single-crossing condition in own action and type; the within-player complementarity of the CDCP satisfies her conditions, but the across-player substitutes violate single-crossing in rivals’ actions, requiring the additional bounding argument of Proposition 8.

Combinatorial discrete choice. Jia (2008) studies store location with strategic substitutes; Arkolakis et al. (2023) formalize the general CDCP framework; Antràs et al. (2017) prove supermodularity under constant CES markups when $(\sigma - 1)/\theta > 1$. We extend their supermodularity result to endogenous variable markups, where activating an additional site changes market shares and hence all firms’ markups—introducing a feedback channel absent under constant markups. The proof shows that this feedback is bounded by the same $(\sigma - 1)/\theta > 1$ condition (Proposition 6).

Variable markups and multinational production. The application draws on the variable markup literature (Atkeson and Burstein 2008; Edmond et al. 2015; Gaubert and Itskhoki 2021), which takes the set of active firms as given, and the multinational production literature (Tintelnot 2017; Antràs et al. 2026; Castro-Vincenzi 2024), which uses constant CES markups. By closing the loop between activation and pricing, we introduce the three-way circularity (activation $\rightarrow$ market shares $\rightarrow$ markups $\rightarrow$ activation incentives) and show that the pricing contraction bounds this feedback, preserving both supermodularity and equilibrium uniqueness. The static game provides the stage-game structure for dynamic oligopoly models of the type studied by Ericson and Pakes (1995) and Ifrach and Weintraub (2017). In the estimation of such dynamic games, Aguirregabiria and Mira (2007) develop sequential methods that require equilibrium selection at each iteration, while Bajari et al. (2007) propose forward-simulation methods that bypass equilibrium compu-

tation altogether; the uniqueness conditions derived here offer a third path, guaranteeing that the static stage game has a single equilibrium to estimate.

2. MODEL

We develop a static model of combinatorial entry with endogenous pricing. Firms choose which production sites to activate across multiple locations and compete in quantities under CES demand with endogenous variable markups. The leading interpretation is multinational production in segmented international markets, though the game-theoretic results apply more broadly. Figure 1 summarizes the three-way circularity and its relationship to the existing literature.

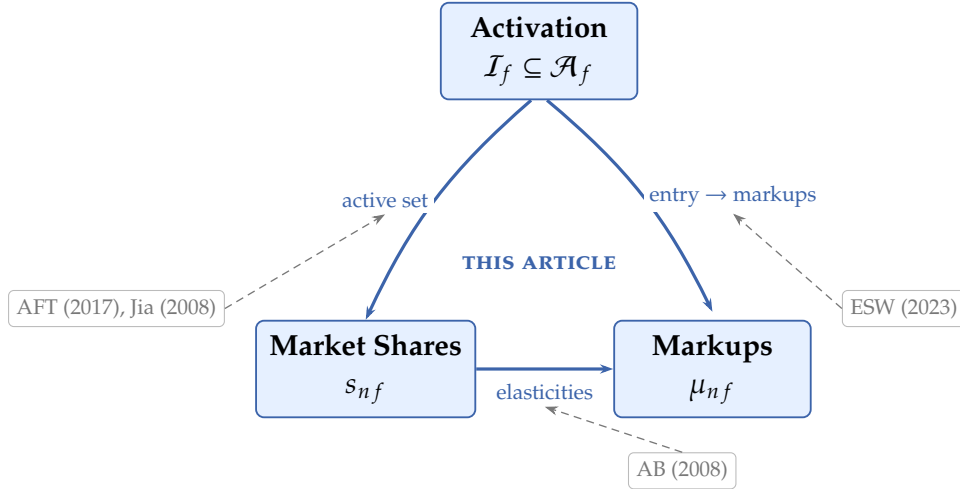


Figure 1: Three-way circularity and the literature. Solid arrows show the full equilibrium loop closed in this article. Prior work addresses subsets: [Atkeson and Burstein \(2008\)](#) endogenize markups given active firms (bottom edge); [Antràs et al. \(2017\)](#) and [Jia \(2008\)](#) solve activation under constant markups (left edge); [Espin-Sanchez et al. \(2023\)](#) prove uniqueness for binary entry decisions given a fixed payoff function (right edge; their result covers the activation channel but does not close the pricing loop).

2.1 Environment

There are N countries, indexed by $n \in \{1, \dots, N\}$. Each country serves as both a potential production site and a destination market. Wages w_n are exogenous.

There are F firms, indexed by $f \in \mathcal{F} = \{1, \dots, F\}$, that produce differentiated varieties in a single sector. Each firm f is characterized by two primitives:

- (i) a *total factor productivity* (TFP) $a_f > 0$, and
- (ii) a *set of available production sites* $\mathcal{A}_f \subseteq \{1, \dots, N\}$.

The firm's strategic decision is which sites in $\mathcal{A}_f$ to activate for production. Section 3.2 extends the model by introducing private information over site-specific fixed costs, yielding a Bayesian game.

2.2 Demand

A representative household in each country has income $E_n > 0$ and nested CES preferences. Following Atkeson and Burstein (2008), there is a continuum of sectors indexed by $k \in [0, 1]$, aggregated with elasticity of substitution $\eta > 1$:

$$C_n = \left[\int_0^1 C_{nk}^{\frac{\eta-1}{\eta}} dk \right]^{\frac{\eta}{\eta-1}}. \quad (2.1)$$

The F oligopolistic firms compete in a single differentiated-good sector (say $k = 0$). Because this sector has measure zero in the continuum, its price index P_n does not affect the aggregate price index $\tilde{P}_n$, so the demand shifter Ω_n defined below is truly exogenous to the firms' decisions. Within the differentiated sector, varieties are aggregated through a CES technology with elasticity $\sigma > \eta$:

$$Q_n = \left[\sum_{f \in \mathcal{F}_n} q_{nf}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad (2.2)$$

where $\mathcal{F}_n$ is the set of firms actively supplying market n . Since all production sites can serve all markets via iceberg trade costs, $f \in \mathcal{F}_n$ if and only if $\mathcal{I}_f \neq \emptyset$: a firm is present in every market once it activates at least one site. The within-sector price index is:

$$P_n = \left[\sum_{f \in \mathcal{F}_n} p_{nf}^{1-\sigma} \right]^{\frac{1}{1-\sigma}}. \quad (2.3)$$

Two-stage budgeting implies that total expenditure on the differentiated sector in

market n is:

$$R_n = \Omega_n P_n^{1-\eta}, \quad (2.4)$$

where $\Omega_n \equiv E_n / \tilde{P}_n^{1-\eta} > 0$ collects the household's income and the aggregate price index $\tilde{P}_n$. Because the differentiated sector is measure zero in the continuum, $\tilde{P}_n$ is determined by the remaining sectors and is therefore exogenous to the oligopolistic firms. Firm f 's demand in market n is:

$$q_{nf} = p_{nf}^{-\sigma} P_n^{\sigma-\eta} \Omega_n. \quad (2.5)$$

Assumption 1 (Fixed sectoral expenditure). Sectoral expenditure $R_n = \Omega_n P_n^{1-\eta}$ is treated as exogenous: we fix R_n at its initial value and suppress the price-index feedback $P_n \rightarrow R_n$.

This assumption is maintained throughout the theoretical analysis and the quantitative illustration. It is expected to be conservative for uniqueness: the omitted channel ($P_n^{1-\eta}$ amplifying demand responses through R_n) would strengthen pricing feedback and tighten the contraction condition, though a formal proof would require bounding the additional $P_n \rightarrow R_n$ loop jointly with the pricing contraction. Extending to fully endogenous R_n would add a sector-level general-equilibrium loop that does not change the structure of the contraction condition but would require additional analysis. The Jia–Tarski gap of zero at all tested parameterizations computationally certifies that the equilibrium under fixed R_n satisfies supermodularity.

2.3 Production Technology and Costs

Firms produce using labor with heterogeneous productivity across production sites. Firm f with TFP a_f delivering one unit of output from production site i to market n faces unit cost:

$$c_{fin}(\omega) = \frac{w_i \tau_{in}}{a_f z(\omega)}, \quad (2.6)$$

where w_i is the wage at site i , $\tau_{in} \geq 1$ is the iceberg trade cost from site i to market n , and $z(\omega)$ is an i.i.d. Fréchet productivity draw with shape parameter $\theta > 0$ across varieties $\omega \in [0, 1]$.

A firm producing at multiple sites sources each variety from the lowest-cost site. As shown by [Eaton and Kortum \(2002\)](#)—and applied at the firm level by [Tintelnot \(2017\)](#) and [Antràs et al. \(2017\)](#)—the Fréchet assumption implies that the firm-level cost index

aggregates across active production sites:

$$c_{nf} = \frac{\Gamma_\theta}{a_f} \left[\sum_{i \in \mathcal{I}_f} (w_i \tau_{in})^{-\theta} \right]^{-\frac{1}{\theta}}, \quad (2.7)$$

where $\mathcal{I}_f \subseteq \mathcal{A}_f$ is the set of production sites that firm f activates and $\Gamma_\theta = \Gamma(1 + \frac{1-\sigma}{\theta})^{\frac{1}{1-\sigma}}$ with $\Gamma(\cdot)$ the Gamma function. Higher TFP a_f lowers the cost index uniformly across all markets. Because Γ_θ is common across all firms, we normalize $\Gamma_\theta = 1$ without loss of generality; it can be absorbed into the TFP parameters $\{a_f\}$.

Remark 2 (Cost Complementarity). Equation (2.7) embodies a key complementarity: activating an additional production site i' adds a term $(w_{i'} \tau_{i'n})^{-\theta}$ to the sum inside the brackets, *reducing* the cost index c_{nf} in every market n . This cost reduction is larger when the firm already has fewer active sites—a form of increasing differences in the activation indicators. [Antràs et al. \(2017\)](#) establish that this cost complementarity generates supermodularity in the global sourcing problem under constant CES markups when $(\sigma - 1)/\theta > 1$.

2.4 Pricing and Markups

Firms compete à la Cournot: each firm chooses the quantity q_{nf} of its variety to supply to market n , taking rivals' quantities as given. The nested CES demand system with a finite number of firms generates endogenous variable markups through a mechanism first analyzed by [Atkeson and Burstein \(2008\)](#).

Firm f 's revenue share in market n is:

$$s_{nf} \equiv \frac{p_{nf}^{1-\sigma}}{P_n^{1-\sigma}} = \frac{p_{nf} q_{nf}}{\sum_{f'} p_{nf'} q_{nf'}}. \quad (2.8)$$

Under Cournot competition, the firm internalizes its effect on the sectoral aggregate Q_n , yielding a *perceived inverse elasticity* of demand (see [Atkeson and Burstein, 2008](#), Section II):

$$\frac{1}{\epsilon_{nf}} = \frac{1 - s_{nf}}{\sigma} + \frac{s_{nf}}{\eta}. \quad (2.9)$$

The first-order condition $p_{nf}(1 - 1/\epsilon_{nf}) = c_{nf}$ yields the optimal price as a markup over marginal cost:

$$p_{nf} = \mu_{nf} \cdot c_{nf}, \quad \mu_{nf} \equiv \frac{\epsilon_{nf}}{\epsilon_{nf} - 1}. \quad (2.10)$$

Remark 3 (Bounded Markups and the Cannibalization Channel). Equation (2.9) implies that the perceived inverse elasticity is a harmonic-mean weight between the two CES parameters σ and η :

- (i) **Small firm** ($s_{nf} \rightarrow 0$): $\epsilon_{nf} \rightarrow \sigma$, $\mu_{nf} \rightarrow \sigma/(\sigma - 1)$. The firm faces the within-sector CES markup.
- (ii) **Dominant firm** ($s_{nf} \rightarrow 1$): $\epsilon_{nf} \rightarrow \eta$, $\mu_{nf} \rightarrow \eta/(\eta - 1)$. The firm's pricing power is bounded by across-sector substitution.

Markups are bounded: $\mu_{nf} \in [\sigma/(\sigma - 1), \eta/(\eta - 1)]$. Activating additional production sites lowers marginal cost (2.7), which raises output and market share, thereby *compressing* the perceived elasticity and *raising* the markup. But the markup is bounded above by $\eta/(\eta - 1)$, so the self-cannibalization force is finite. This *bounded* cannibalization—a direct consequence of the nested CES structure (Atkeson and Burstein, 2008)—competes against the cost complementarity of Remark 2. Whether supermodularity of the activation problem survives this tension is the central question of Section 3. A second, cross-firm channel arises because activation changes the rival demand index (2.11), feeding back into all firms' markups—a pricing interaction absent under constant markups that creates a new obstacle to uniqueness (Proposition 11).

Remark 4 (Cournot vs. Bertrand). Under Cournot, the perceived inverse elasticity is $1/\epsilon^C = (1 - s)/\sigma + s/\eta$ (equation 2.9). Under Bertrand with the same nested CES demand, the perceived elasticity is $\epsilon^B = \sigma(1 - s) + \eta s$, which differs for interior market shares: the two coincide only at $s = 0$ (both yield σ) and $s = 1$ (both yield η). The theoretical framework—pricing contraction via the interaction matrix, supermodularity via cost-aggregator convexity—applies to both cases, but the quantitative conditions (the feedback coefficient κ , the per-market threshold, and the contraction modulus $\gamma_{\max}$) must be re-derived using the Bertrand markup function $\mu^B(s) = \epsilon^B/(\epsilon^B - 1)$. Since Bertrand markups are lower than Cournot markups at interior shares, the pricing contraction is typically weaker, but this must be verified case by case. We adopt the Cournot assumption

throughout, which is natural for the multinational production application where firms choose production scale and capacity.

Rival demand index. The competitive environment facing firm f in market n is summarized by a *rival demand index*:

$$D_{-f}(n) = \sum_{f' \neq f, f' \in \mathcal{F}_n} p_{nf'}^{1-\sigma}, \quad (2.11)$$

which is a sufficient statistic for competition: firm f needs only $D_{-f}(n)$ to compute its optimal price and variable profit in market n . The firm's market share can be written as:

$$s_{nf} = \frac{p_{nf}^{1-\sigma}}{p_{nf}^{1-\sigma} + D_{-f}(n)}, \quad (2.12)$$

making explicit the dependence of the markup (2.9) on the rival demand index.

2.5 The Activation Problem

Each firm f chooses which production sites to activate. Let $I_{fi} \in \{0, 1\}$ be an indicator equal to 1 if firm f activates production site $i \in \mathcal{A}_f$, and 0 otherwise. The firm's *activation set* is:

$$\mathcal{I}_f = \{i \in \mathcal{A}_f : I_{fi} = 1\}. \quad (2.13)$$

Activating production site i incurs a fixed operating cost $\phi_i \cdot w_i$ in units of local labor, where ϕ_i may vary across countries (reflecting, e.g., differences in regulatory or infrastructure costs).

Firm f 's total static profit, given its own activation set $\mathcal{I}_f$ and the equilibrium activation sets of all rivals $\mathcal{I}_{-f}^*$, is:

$$\pi_f(\mathcal{I}_f; \mathcal{I}_{-f}^*) = \sum_{n=1}^N \left[\text{rev}_{nf}(\mathcal{I}_f, \mathcal{I}_{-f}^*) \cdot \frac{\mu_{nf} - 1}{\mu_{nf}} \right] - \sum_{i \in \mathcal{I}_f} \phi_i w_i, \quad (2.14)$$

where $\text{rev}_{nf} = p_{nf} \cdot q_{nf}$ is firm f 's revenue in market n and $(\mu_{nf} - 1)/\mu_{nf}$ is the variable profit margin. Revenue and markups depend on the full industry activation profile: own

activation enters through the cost index (2.7), and rivals' activation sets enter through the rival demand index $D_{-f}(n)$ defined in (2.11).

2.6 Static Equilibrium

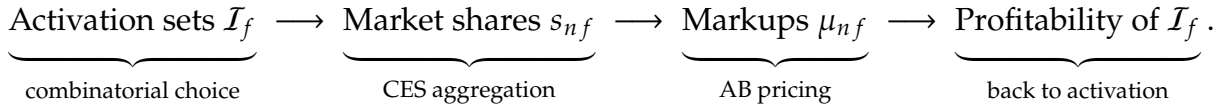
Definition 5 (Static Equilibrium). A *static equilibrium* is a profile $(\{\mathcal{I}_f^*\}_{f \in \mathcal{F}}, \{\mu_{nf}^*\}, \{s_{nf}^*\})$ of activation sets, markups, and market shares such that:

E1. Profit maximization: For each firm f , the activation set $\mathcal{I}_f^*$ solves

$$\mathcal{I}_f^* \in \arg \max_{\mathcal{I}_f \subseteq \mathcal{A}_f} \pi_f(\mathcal{I}_f; \mathcal{I}_{-f}^*). \quad (2.15)$$

E2. Markup consistency: For each (n, f) with $\mathcal{I}_f^* \neq \emptyset$, the markup μ_{nf}^* satisfies the pricing equation (2.10) with perceived elasticity (2.9), and market shares s_{nf}^* are consistent with equilibrium prices via (2.12).

The static equilibrium involves a three-way circularity:



With constant markups (as in Antràs et al., 2017), the last link is severed: activation does not feed back through markups. The theoretical contribution of this article is to show that the CDCP structure—and in particular, supermodularity—survives this feedback (Section 3). A complementary computational contribution compares alternative algorithms for resolving the three-way fixed point (Section 4).

3. THEORETICAL RESULTS

This section establishes the theoretical properties of the static game. We first show that each firm's activation problem is supermodular (Proposition 6), then characterize the equilibrium set (Proposition 8) and establish convergence (Proposition 9). The main result—BNE uniqueness via a Jacobian decomposition—builds on the pricing-uniqueness condition of Section 3.1 and is stated formally in Section 3.2.

Since the rival demand index $D_{-f}(n)$ is a sufficient statistic for competition (Section 2.4), we write $\pi_f(\mathcal{I}_f; D_{-f})$ for firm f 's profit (2.14) evaluated at the unique pricing equilibrium given D_{-f} .

Proposition 6 (Single-Agent Supermodularity). *Fix the rival demand index $D_{-f}(n)$ for all markets n and suppose $(\sigma - 1)/\theta > 1$.*

- (i) *Per-market sufficient condition (CES benchmark). If the equilibrium market shares satisfy $\max_{n,f} s_{nf} < \bar{s}^*$, where $\bar{s}^* \equiv (\alpha - 1)/(2\alpha)$ with $\alpha = (\sigma - 1)/\theta$ is a tractable conservative threshold derived under constant CES markups (Appendix A, Step 4), then each market's contribution to variable profit is convex in the cost aggregator, which implies that firm f 's activation profit $\pi_f(\mathcal{I}_f; D_{-f})$ exhibits increasing differences in the activation indicators $\{I_{fi}\}_{i \in \mathcal{A}_f}$. Under AB endogenous markups, the exact per-market condition is given implicitly by $H_{AB}(s) \geq 0$ (Appendix A, Step 3'):*

$$\pi_f(\mathcal{I}_f \cup \{i, i'\}; D_{-f}) - \pi_f(\mathcal{I}_f \cup \{i\}; D_{-f}) \geq \pi_f(\mathcal{I}_f \cup \{i'\}; D_{-f}) - \pi_f(\mathcal{I}_f; D_{-f}). \quad (3.1)$$

- (ii) *Computational certification. When some market shares exceed $\bar{s}^*$, the per-market condition is not met, but aggregate supermodularity across all markets can still hold because low-share markets (positive curvature) dominate high-share markets (negative curvature). The Jia–Tarski algorithm certifies aggregate supermodularity whenever the upper and lower bounds coincide (gap = 0).*

Under either (i) or (ii), the set of optimal solutions to $\max_{\mathcal{I}_f \subseteq \mathcal{A}_f} \pi_f(\mathcal{I}_f; D_{-f})$ has a greatest element and a least element, each attainable by the Jia–Tarski algorithm.

Proof sketch. The marginal return to activating site i' can be decomposed into two effects. The *cost complementarity effect*: by equation (2.7), adding site i' contributes a term $(w_{i'} \tau_{i'n})^{-\theta}$ to the cost index in every market n . The proportional cost reduction is larger when the existing cost index is higher—i.e., when fewer other sites are already active. Under the condition $(\sigma - 1)/\theta > 1$, variable profit is sufficiently convex in the cost index that the scale expansion from lower costs generates increasing differences.

The *self-cannibalization effect*: activating i' lowers cost, which lowers price, which raises market share s_{nf} , which compresses the perceived elasticity ϵ_{nf} via (2.9) and

raises the markup μ_{nf} . This partially offsets the quantity gain. Under the CES benchmark (constant markups), per-market convexity of variable profit in the cost aggregator $\Sigma_{fn} \equiv \sum_{i \in \mathcal{I}_f} (w_i \tau_{in})^{-\theta}$ from (2.7) requires $s < \bar{s}^*$. Under AB endogenous markups, the exact per-market condition is $H_{AB}(s) \geq 0$, where H_{AB} is an explicit function of $(s, \sigma, \eta, \theta)$ derived in Appendix A, Step 3'; the AB threshold is parameter-dependent and need not exceed the CES threshold. When some shares exceed both thresholds, aggregate supermodularity is verified via part (ii).

The condition $(\sigma - 1)/\theta > 1$ is the same as in Antràs et al. (2017), Theorem 1, who establish supermodularity under constant markups. The extension here shows that the additional markup feedback can be incorporated analytically at the per-market level and does not overturn supermodularity in the tested parameterizations. Full proof in Appendix A. $\square$

Under Atkeson–Burstein endogenous markups, the exact per-market convexity condition $H_{AB}(s) \geq 0$ defines a *unique* implicit threshold $\bar{s}_{AB}^*$ (Appendix A, Proposition 22): H_{AB} is strictly concave, so the threshold is the unique zero of a concave function that is positive at $s = 0$ and negative at $s = 1$. The AB threshold is parameter-dependent and need not uniformly exceed the CES benchmark $\bar{s}^*$. In the cloud application (Section 5), $\max_{n,f} s_{nf} = 0.40$ exceeds both thresholds, so aggregate supermodularity is certified computationally via part (ii): the Jia–Tarski gap is zero at all iterations.

Remark 7 (Supermodularity in practice). Propositions 8–15 require only that the single-agent activation problem is supermodular—whether certified analytically via part (i) or computationally via part (ii) is immaterial to the argument. Note that the per-market condition in part (i) must hold across *all* possible activation sets $\mathcal{I}_f$, not only the equilibrium one, because the market share s_{nf} depends on the cost index $\Sigma_{fn}(\mathcal{I}_f)$ which varies with the activation set. The computational certification in part (ii) verifies supermodularity at all relevant activation sets via the Jia–Tarski algorithm and is the recommended diagnostic.

Existence. A Nash equilibrium of the complete-information game exists by Nash's theorem, since the activation lattice $L = \prod_f 2^{\mathcal{A}_f}$ is finite. The following two propositions characterize the equilibrium set and establish convergence of iterative algorithms.

Proposition 8 (Equilibrium Bounds). *Suppose each firm's activation profit is supermodular (Proposition 6).*

- (i) The joint best-response map Φ is anti-monotone (order-reversing) on the joint activation lattice L , so the squared map $\Phi^2 = \Phi \circ \Phi$ is monotone. By Tarski's (1955) fixed-point theorem, Φ^2 has a greatest fixed point $\bar{\mathbf{I}}$ and a least fixed point $\underline{\mathbf{I}}$ on L .
- (ii) Any Nash equilibrium $\mathbf{I}^*$ satisfies $\underline{\mathbf{I}} \leq \mathbf{I}^* \leq \bar{\mathbf{I}}$: the extremal fixed points of Φ^2 bound the entire Nash equilibrium set.
- (iii) If $\bar{\mathbf{I}}$ is itself a Nash equilibrium (i.e., $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$, not merely a two-cycle of Φ), it is the greatest Nash equilibrium.

Proof. Since activation choices are *strategic substitutes*—when rival firms activate more sites, the rival demand index D_{-f} increases, reducing the profitability of firm f 's sites—the joint best-response map $\Phi : (I_1, \dots, I_F) \mapsto (I_1^*, \dots, I_F^*)$ is *anti-monotone*: $\mathbf{I} \leq \mathbf{I}'$ implies $\Phi(\mathbf{I}) \geq \Phi(\mathbf{I}')$ (cf. Amir, 1996), where $\leq$ is the component-wise set-inclusion order on $L \equiv \prod_f 2^{\mathcal{A}_f}$.

Part (i): Φ^2 is monotone because anti-monotonicity composes to give $\mathbf{I} \leq \mathbf{I}' \Rightarrow \Phi^2(\mathbf{I}) \leq \Phi^2(\mathbf{I}')$. Tarski's theorem on the finite complete lattice L yields the extremal fixed points.

Part (ii): any Nash equilibrium satisfies $\Phi(\mathbf{I}^*) = \mathbf{I}^*$, hence $\Phi^2(\mathbf{I}^*) = \mathbf{I}^*$. By Tarski's theorem, $\underline{\mathbf{I}} \leq \mathbf{I}^* \leq \bar{\mathbf{I}}$.

Part (iii): if $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$, then $\bar{\mathbf{I}}$ is a Nash equilibrium; any other Nash equilibrium $\mathbf{I}^*$ satisfies $\mathbf{I}^* \leq \bar{\mathbf{I}}$ by (ii).

In general, $\bar{\mathbf{I}}$ may be a two-cycle rather than a Nash equilibrium: it is possible that $\Phi(\bar{\mathbf{I}}) \neq \bar{\mathbf{I}}$ while $\Phi^2(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$. However, at all parameterizations tested in Section 5 and Appendix F, $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$ and the equilibrium is unique. Full proof in Appendix B. $\square$

Proposition 9 (Convergence of Jacobi Iteration). *Suppose each firm's activation profit is supermodular. Starting from the supremum $\hat{\mathbf{I}} = (\mathcal{A}_1, \dots, \mathcal{A}_F)$, the even subsequence $\{\Phi^{2k}(\hat{\mathbf{I}})\}$ is decreasing and converges to $\bar{\mathbf{I}}$ (the greatest fixed point of Φ^2). The odd subsequence $\{\Phi^{2k+1}(\hat{\mathbf{I}})\}$ is increasing and converges to $\Phi(\bar{\mathbf{I}})$. If $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$, the full sequence converges to the greatest Nash equilibrium. Analogously from the infimum.*

Proof. Since Φ is anti-monotone and $\Phi(\hat{\mathbf{I}}) \leq \hat{\mathbf{I}}$, the even subsequence $\{\Phi^{2k}(\hat{\mathbf{I}})\}$ is decreasing (by monotonicity of Φ^2) and the odd subsequence $\{\Phi^{2k+1}(\hat{\mathbf{I}})\}$ is increasing, with $\Phi^{2k+1}(\hat{\mathbf{I}}) \leq \Phi^{2k}(\hat{\mathbf{I}})$ for all k . On the finite lattice L , both subsequences stabilize in finitely many steps. The even limit is $\bar{\mathbf{I}}$ and the odd limit is $\Phi(\bar{\mathbf{I}})$, which satisfy $\Phi^2(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$ and $\Phi(\bar{\mathbf{I}}) \leq \bar{\mathbf{I}}$. Full proof in Appendix B. $\square$

Remark 10 (Gauss–Seidel convergence). The sequential update map Ψ that updates firms in order $(f_1, \dots, f_F)$ using the latest available rivals’ strategies converges at all tested parameterizations (Section 5 and Appendix F). Formal convergence of Ψ under anti-monotone best responses is not guaranteed in general—sequential updating breaks the even/odd subsequence structure that yields monotonicity of Φ^2 —but empirical convergence is robust across all parameterizations tested.

At all parameterizations tested in Section 5 and Appendix F, $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$: both Jacobi and Gauss–Seidel converge to the same Nash equilibrium, and no two-cycles are observed. Gauss–Seidel uses updated rival information within each round and typically converges in fewer rounds, whereas Jacobi is naturally parallelizable. Section 4 presents the full comparison.

3.1 Pricing Uniqueness and the Intensive Margin

The uniqueness question has two distinct margins. On the *intensive margin* (pricing), uniqueness requires that the AB markup fixed point is contractive for any given activation profile. On the *extensive margin* (site activation), uniqueness requires that no “marginal sites”—sites whose profitability depends on rival behavior—switch activation status across equilibria. This subsection resolves the intensive margin via a contraction condition on the feedback coefficient (Proposition 11); Section 3.2 resolves the extensive margin by introducing private information over fixed costs.

Feedback coefficient. The strength of the strategic interaction through the AB markup channel is captured by a *feedback coefficient*. For firm f in market n with share s_{nf} and markup μ_{nf} , define:

$$\kappa_{nf} = \frac{s_{nf}^2 \beta_{nf}}{1 + s_{nf}(1 - s_{nf}) \beta_{nf}}, \quad \beta_{nf} = \frac{(\sigma - 1) \mu_{nf} (\sigma - \eta)}{\sigma \eta}. \quad (3.2)$$

The coefficient $\kappa_{nf} = \partial p_{nf}^{1-\sigma} / \partial D_{-f}(n)$ measures how firm f ’s pricing power responds to a change in the rival demand index, holding the activation set fixed. When $s_{nf} = 0$ (negligible firm), $\kappa_{nf} = 0$; as s_{nf} rises, the markup channel amplifies the pricing response, so κ_{nf} increases.

Proposition 11 (Pricing Uniqueness). *For any fixed activation profile $(\mathcal{I}_1, \dots, \mathcal{I}_F)$, define the markup interaction matrix Λ with entries*

$$\Lambda_{(n,f),(n,g)} = \kappa_{nf} \cdot \frac{p_{ng}^{1-\sigma}}{D_{-f}(n)}, \quad g \neq f, \quad (3.3)$$

with all other entries zero. If the spectral radius satisfies

$$\rho(\Lambda) < 1, \quad (3.4)$$

then the AB pricing equilibrium is unique.

Proof. For a fixed activation profile, the AB pricing system defines a map $\Psi : D \mapsto D'$ on the space of rival demand indices. The Jacobian of Ψ is bounded entrywise by Λ , which is nonnegative. When $\rho(\Lambda) < 1$, the Perron–Frobenius theorem guarantees a positive vector $w \gg 0$ and a scalar $\lambda < 1$ such that $\Lambda w \leq \lambda w$. Define the weighted sup-norm $\|x\|_w = \max_i |x_i|/w_i$. By the mean value theorem, for any two demand vectors D, D' :

$$\|\Psi(D) - \Psi(D')\|_w \leq \lambda \|D - D'\|_w,$$

so Ψ is a contraction in $\|\cdot\|_w$. Since the domain is a compact convex subset of $\mathbb{R}_+^{NF}$ (bounded markups imply bounded demand indices), Banach’s fixed-point theorem yields a unique pricing equilibrium. $\square$

The matrix Λ is block-diagonal across markets (N blocks of size $F \times F$), so $\rho(\Lambda) = \max_n \rho(\Lambda_n)$ is cheap to compute. A conservative closed-form sufficient condition uses the infinity norm:

Corollary 12. *If $\max_{n,f} \kappa_{nf} < 1/(F - 1)$, then $\rho(\Lambda) < 1$ and the pricing equilibrium is unique.*

At the cloud application parameterization ($F = 5$, Section 5), both conditions are satisfied: $\max \kappa = 0.193 < 1/(F - 1) = 0.25$ and $\rho(\Lambda) = 0.103$. At the synthetic economy ($F = 8$, Appendix F), the infinity-norm condition is violated ($\max \kappa = 0.193 > 1/(F - 1) = 0.143$) while the spectral condition holds ($\rho(\Lambda) = 0.096$). For applied work, the spectral radius (3.4) is the recommended diagnostic.

Proposition 11 resolves the intensive margin but does *not* guarantee uniqueness of the full game, because different activation profiles lead to different demand indices. Define the

composite map $\tilde{\Phi} : D \mapsto D'$ that maps rival demands to best-response activation sets, then evaluates the unique pricing equilibrium to produce new demands. Multiple equilibria arise when $\tilde{\Phi}$ has multiple fixed points—equivalently, when there exist marginal sites whose activation profit $\Delta\pi_{if}$ changes sign across equilibrium demands. The number of marginal sites is the key diagnostic for multiplicity; the cloud application (Section 5.5) and synthetic stress tests (Appendix F) show it is zero at all tested parameterizations. To resolve the extensive margin analytically, the next subsection introduces private information over fixed costs, converting the discrete activation problem into a continuous threshold system on which contraction arguments apply.

3.2 Bayesian Extension with Private Costs

In practice, firms' fixed costs are not publicly observed. We model this by introducing private information over fixed costs, yielding a Bayesian game whose equilibrium is characterized by a threshold system on which contraction-based uniqueness arguments can be applied.

Assumption 13 (Private cost shocks). Each firm f privately observes fixed-cost shocks $\{\varepsilon_{if}\}_{i \in \mathcal{A}_f}$ that are i.i.d. across sites and across firms, drawn from a distribution G with full-support density g , mean zero, and scale parameter $\sigma_\varepsilon > 0$. The total fixed cost of activating site i is $(\phi_i + \varepsilon_{if})w_i$. Firm f observes its own draws but not rivals'.

The i.i.d. assumption is substantive: it ensures that each site's activation decision reduces to a scalar threshold comparison (Proposition 15(i)), enabling the Jacobian factorization (3.8). With correlated shocks—for example, a hierarchical structure $\varepsilon_{if} = \eta_f + \tilde{\varepsilon}_{if}$ with a common firm-level component η_f —the activation decision for site i would depend on η_f through the conditional expected marginal value, yielding a threshold that varies with the firm-level draw rather than an unconditional scalar. The Jacobian factorization breaks because the density factor becomes conditional on η_f . A partial extension is possible: conditional on η_f , the site-level shocks $\tilde{\varepsilon}_{if}$ remain i.i.d., so a conditional version of the contraction argument applies at each realization, but the fixed point is then over a higher-dimensional system (one threshold per site per firm-level type) and the contraction condition is harder to verify. In practice, the i.i.d. assumption is standard in the CDCP

literature (Jia, 2008; Arkolakis et al., 2023); the restriction here is no more binding than in those frameworks.

Economically, the i.i.d. assumption rules out two forms of dependence that arise naturally in multinational production. First, *within-firm correlation*: a firm-level “headquarters quality” draw that shifts fixed costs at all sites simultaneously—for example, a management practice that raises or lowers setup costs uniformly across countries. Second, *spatial correlation*: regulatory or institutional costs that co-move across neighboring countries sharing a border or regulatory regime. Either form of dependence would break the one-dimensional threshold structure exploited in Proposition 15(i).

Two distinct roles of the i.i.d. assumption deserve emphasis. Independence across sites *within* a given firm is standard in the CDCP literature: it ensures that each site’s activation reduces to a scalar threshold comparison, enabling the Jia–Tarski algorithm. Independence *across firms* does additional work here: it makes each firm’s threshold an unconditional scalar that does not depend on other firms’ cost realizations, so the cross-player Jacobian J_T has the product structure (3.8) that supports the contraction argument. With correlated shocks across firms—for example, a common macroeconomic shock—the thresholds would be conditional on the common state, and the contraction condition would need to hold state-by-state.

Extension: across-firm common shocks. The state-by-state requirement is easy to satisfy in the paper’s framework. Consider a public-state decomposition

$$\varepsilon_{if} = \zeta + \tilde{\varepsilon}_{if}, \quad \zeta \in \mathbb{R}, \quad \tilde{\varepsilon}_{if} \stackrel{\text{i.i.d.}}{\sim} G, \quad (3.5)$$

where ζ is a common shock observed by all firms (e.g., a cost-of-capital shock, a country-cluster regulatory tightening) and $\tilde{\varepsilon}_{if}$ are residual idiosyncratic draws satisfying Assumption 13. Conditional on ζ , the game coincides with the paper’s baseline model after the deterministic relabeling $\phi_i \mapsto \phi_i + \zeta$: the threshold map, its Jacobian, and the contraction modulus $\gamma(\sigma_\varepsilon \mid \zeta)$ are the same objects evaluated at shifted mean fixed costs. The following proposition records the consequence.

Proposition 14 (BNE uniqueness under across-firm common shock). *Under the decomposi-*

tion (3.5), the BNE exists and is unique state-by-state whenever

$$\sup_{\zeta \in \text{supp}(\zeta)} \gamma_{\max}(\zeta) < 1, \quad (3.6)$$

where $\gamma_{\max}(\zeta) \equiv \max_{\sigma_\varepsilon > 0} \gamma(\sigma_\varepsilon \mid \zeta)$ is the bathtub peak evaluated at shifted fixed costs $(\phi_i + \zeta)_{i=1}^N$.

Proof. Conditional on ζ , the game is Assumption 13 applied to $\tilde{\varepsilon}_{if}$ with deterministic fixed costs $(\phi_i + \zeta)w_i$. Propositions 6–15 and Theorem 18 apply unchanged, so BNE uniqueness at state ζ follows from $\gamma_{\max}(\zeta) < 1$. Taking the supremum over the support of ζ delivers state-by-state uniqueness with a single primitive-level condition. $\square$ $\square$

Condition (3.6) is verifiable by evaluating the bathtub peak on a grid of ζ values, or—more directly—by checking BNE uniqueness state-by-state via the Jia–Tarski three-starting-point test. Table 1 reports the latter for $\zeta/\bar{\phi} \in \{-0.4, -0.2, -0.1, 0, +0.1, +0.2, +0.4\}$ in the small economy of Section 3.4—a $\pm 40\%$ shift in mean fixed costs covers approximately two standard deviations of any reasonable common shock. At three values of σ_ε bracketing the bathtub peak, the BNE fixed-point iteration from supremum, infimum, and uniform starting points converges to the same activation probability vector at every ζ , with maximum pairwise distances below 10^{-3} . By Proposition 14, uniqueness extends to the full game integrating over ζ .

The remaining case—*within-firm* common shocks $\varepsilon_{if} = \eta_f + \tilde{\varepsilon}_{if}$ with η_f privately observed by firm f —is genuinely outside the paper’s framework: it breaks the sibling independence used to reduce each firm’s CDCP to $|\mathcal{A}_f|$ scalar threshold comparisons. A formal treatment would condition the threshold system on firm-level types and replace Banach contraction with a family of contractions indexed by η_f ; we leave this extension to future work. Within-firm correlation is arguably the less empirically relevant case in our application, because calibrated site-specific fixed costs ϕ_i already absorb the firm-invariant component of country-site heterogeneity.

The solution concept is Bayesian Nash equilibrium (BNE). Existence is immediate: the threshold map $T : [0, 1]^M \rightarrow [0, 1]^M$ (defined below) is continuous, so Brouwer’s fixed-point theorem guarantees a BNE in threshold strategies.

Threshold strategies. The single-agent activation problem is supermodular in $(\mathcal{I}_f, -\varepsilon_f)$: higher fixed-cost shocks make activation less attractive, and the interaction is comple-

Table 1: Tier T2: Across-firm common-shock robustness

$\zeta/\bar{\phi}$	ζ	$\max \ p_{\text{sup}} - p_{\text{inf}}\ _{\infty} \text{ at } \sigma_{\varepsilon}/\bar{\phi} =$			BNE unique?
		0.10	0.30	1.00	
-0.40	-0.0493	5.3e-08	7.1e-08	2.9e-07	yes
-0.20	-0.0247	7.3e-09	1.8e-07	2.8e-07	yes
-0.10	-0.0123	1.1e-10	2.9e-07	2.6e-07	yes
+0.00	+0.0000	6.1e-13	1.6e-07	2.5e-07	yes
+0.10	+0.0123	2.7e-15	1.8e-07	2.4e-07	yes
+0.20	+0.0247	2.0e-18	8.2e-08	2.2e-07	yes
+0.40	+0.0493	5.7e-26	2.5e-08	1.8e-07	yes

Notes: Small-economy calibration ($N = 3, F = 4$; see Section 3.4). For each common-shock realization ζ , the BNE fixed point is solved from three starting points (supremum, infimum, uniform) at three values of σ_{ε} bracketing the bathtub peak. Reported entries are the maximum pairwise $\|p_{\text{sup}} - p_{\text{inf}}\|_{\infty}$ distance across the three starting points. “BNE unique” is yes when the distance is below 10^{-3} at all three σ_{ε} . The $\zeta = 0$ row coincides with the paper’s baseline IID case. Uniqueness holds state-by-state across the full $\pm 40\%$ cost shift, verifying Proposition 14.

mentary by Proposition 6. By the monotone comparative statics of Athey (2001), the BNE involves *threshold strategies*: firm f activates site i if and only if $\varepsilon_{if} < \bar{\varepsilon}_{if}$, where $\bar{\varepsilon}_{if}$ is an endogenous threshold. Given rivals’ thresholds, each site i of rival g is active independently with probability $v_{ig} = G(\bar{\varepsilon}_{ig})$.

Threshold system. The threshold $\bar{\varepsilon}_{if}$ is pinned down by the zero-marginal-value condition: at $\varepsilon_{if} = \bar{\varepsilon}_{if}$, the expected marginal value of activating site i , integrating over rivals’ random activation sets, equals zero. Define the best-response map T that maps rival activation probabilities $\{v_{ig}\}_{g \neq f}$ to own activation probabilities $\{v_{if}\}$ via:

$$\{v_{ig}\}_{g \neq f} \xrightarrow{\text{expected demand}} \mathbb{E}[D_{-f}] \xrightarrow{\rho < 1} \text{pricing eq.} \xrightarrow{\text{bounded markups}} \bar{\varepsilon}_{if} \xrightarrow{G} v_{if}. \quad (3.7)$$

The nested CES demand structure provides three ingredients for this map to be contractive:

1. *Bounded markups* $\mu \in [\sigma/(\sigma-1), \eta/(\eta-1)]$: the marginal value of each site is Lipschitz-continuous in the rival demand index, so small changes in the pricing equilibrium produce bounded changes in the threshold $\bar{\varepsilon}_{if}$, with Lipschitz constant depending on (σ, η) .
2. *Pricing contraction* $\rho(\Lambda) < 1$: the map from expected rival demands to the pricing

equilibrium is contractive (Proposition 11), ensuring that a small shift in activation probabilities feeds back into prices with bounded magnification.

3. *Smoothing via G*: the map from thresholds to probabilities (via the CDF) is smooth and bounded, converting the discrete extensive margin into a continuous probability.

Jacobian decomposition. Before stating the uniqueness result, we develop its key tool: a quantitative bound on the contraction modulus of the threshold map T defined in (3.7). Write $G(x) = G_0(x/\sigma_\varepsilon)$ with standardized CDF G_0 and density $g_0 = G'_0$, so that $g(x) = g_0(x/\sigma_\varepsilon)/\sigma_\varepsilon$. The Jacobian of T factors as

$$J_T(v) = \underbrace{\text{diag}\left(\frac{g_0(\bar{\varepsilon}_{if}/\sigma_\varepsilon)}{\sigma_\varepsilon}\right)}_{\text{density dampening}} \cdot \underbrace{\text{diag}\left(\frac{1}{w_i}\right)}_{\text{scaling}} \cdot \underbrace{\left[\frac{\partial \mathbb{E}[\Delta\pi_{if}]}{\partial v_{jg}}\right]}_{\equiv J_\Delta: \text{marginal-value sensitivity}}, \quad (3.8)$$

where $\bar{\varepsilon}_{if} = \mathbb{E}[\Delta\pi_{if}]/w_i$ is the threshold of site (i, f) . Define $J_v \equiv \text{diag}(1/w_i) \cdot J_\Delta$, which absorbs the wage scaling into the sensitivity matrix. The three ingredients above appear as three factors governing the contraction modulus $\gamma(\sigma_\varepsilon) \equiv \sup_v \|J_T(v)\|_\infty$:

- (a) *Density dampening*: the prefactor $g_0(\cdot)/\sigma_\varepsilon$ vanishes as $\sigma_\varepsilon \rightarrow \infty$.
- (b) *Marginal-value sensitivity*: the matrix J_Δ inherits bounded entries from the CES demand structure. The across-firm entries are non-positive (strategic substitutes through $D_{-f}(n)$), while within-firm entries are non-negative (complementarity from Proposition 6). Both are bounded by the markup range $[\sigma/(\sigma-1), \eta/(\eta-1)]$, and the pricing contraction amplifies demand perturbations by at most $1/(1 - \|\Lambda\|_\infty)$ (Appendix C).
- (c) *Threshold location*: through $g_0(\bar{\varepsilon}_{if}/\sigma_\varepsilon)$, the density is small when the threshold is far from the mode—i.e., when site i is far from marginal.

Proposition 15 (BNE Uniqueness). *Suppose each firm's activation profit is supermodular and the pricing equilibrium is unique ($\rho(\Lambda) < 1$, Proposition 11; a conservative sufficient condition is $\max \kappa < 1/(F-1)$, Corollary 12).*

- (i) *The BNE is in threshold strategies: firm f activates site i iff $\varepsilon_{if} < \bar{\varepsilon}_{if}$.*

(ii) Define the contraction modulus

$$\gamma(\sigma_\varepsilon) \equiv \sup_{v \in [0,1]^M} \|J_T(v)\|_\infty \leq \frac{\|g_0\|_\infty}{\sigma_\varepsilon} \cdot \|J_v\|_\infty, \quad (3.9)$$

where $M = \sum_f |\mathcal{A}_f|$, $\|g_0\|_\infty = \sup_z g_0(z)$ is the peak density of the standardized distribution, and $\|J_v\|_\infty = \sup_v \max_{(i,f)} (1/w_i) \sum_{(j,g)} |\partial \mathbb{E}[\Delta \pi_{if}] / \partial v_{jg}|$ is the weighted row-sum norm of the marginal-value sensitivity. If $\gamma(\sigma_\varepsilon) < 1$, the threshold map T is a contraction and the BNE is unique.

(iii) As $\sigma_\varepsilon \rightarrow 0$, the BNE activation probabilities converge to $\{0, 1\}$, recovering a Nash equilibrium of the complete-information game.

(iv) Assume additionally that G_0 has log-concave tails (e.g., normal or logistic). The map $\sigma_\varepsilon \mapsto \gamma(\sigma_\varepsilon)$ vanishes at both extremes: $\gamma \rightarrow 0$ as $\sigma_\varepsilon \rightarrow \infty$ (density dampening dominates) and $\gamma \rightarrow 0$ as $\sigma_\varepsilon \rightarrow 0$ when all sites are non-marginal (thresholds pushed to density tails). If the complete-information game has a unique Nash equilibrium with gap $\delta_v = \min_{i,f} |\Delta \pi_{if}^*|/w_i > 0$ and $\gamma_{\max} \equiv \max_{\sigma_\varepsilon > 0} \gamma(\sigma_\varepsilon) < 1$, then the BNE is unique for all $\sigma_\varepsilon > 0$. The condition $\gamma_{\max} < 1$ is verifiable numerically by evaluating γ on a grid of σ_ε values.

Proof sketch. Part (i): conditional on rivals' threshold strategies (which determine activation probabilities $\{v_{jg}\}_{g \neq f}$), firm f faces a single-agent CDCP with stochastic fixed costs $(\phi_i + \varepsilon_{if})w_i$. By Proposition 6, this CDCP is supermodular in the activation indicators. Since the shocks ε_{if} are i.i.d. across sites, the optimal activation set is increasing in $-\varepsilon_f$ componentwise on the product lattice $\{0, 1\}^{|\mathcal{A}_f|}$ (Topkis 1998, Theorem 4.2.2). It follows that each site i has a threshold $\bar{\varepsilon}_{if}$ (depending on the equilibrium probabilities of all other sites) such that site i is activated iff $\varepsilon_{if} < \bar{\varepsilon}_{if}$. Crucially, $\bar{\varepsilon}_{if}$ is an *unconditional scalar*: it does not depend on firm f 's own-site shocks ε_{jf} for $j \neq i$. This follows directly from Assumption 13: since $\varepsilon_{if} \perp \varepsilon_{jf}$, the conditional expectation $\mathbb{E}[\Delta \pi_{if} \mid \varepsilon_{if}]$ does not depend on other sites' shock realizations, so the activation decision for site i reduces to a one-dimensional comparison.

For (ii), the factorization (3.8) decomposes each row of $|J_T|$ into density dampening ($\leq \|g_0\|_\infty/\sigma_\varepsilon$) times the weighted marginal-value sensitivity J_v . Since the expected marginal value is a polynomial in $\{v_{ig}\}$, it is smooth; its row sums are bounded by composing

Lipschitz constants along the chain $v \rightarrow \mathbb{E}[D_{-f}] \rightarrow \text{pricing} \rightarrow \Delta\pi_{if}$, with the pricing contraction amplifying perturbations by at most $1/(1 - \rho(\Lambda))$. This yields $\|J_v\|_\infty \leq L_D \cdot L_\mu/(1 - \|\Lambda\|_\infty) + L_C < \infty$ (Appendix C), so $\gamma \leq (\|g_0\|_\infty/\sigma_\varepsilon) \cdot \|J_v\|_\infty < 1$ for σ_ε large enough; Banach’s theorem yields uniqueness.

Part (iii) follows from upper hemicontinuity of the equilibrium correspondence: as $\sigma_\varepsilon \rightarrow 0$, the perturbed game converges to the complete-information game, and any sequence of BNE has a limit point that is a Nash equilibrium of the complete-information game (cf. [Harsanyi 1973](#); for entry games with private information, see [Espin-Sanchez et al. 2023](#), Proposition 2).

For (iv), $\gamma \rightarrow 0$ as $\sigma_\varepsilon \rightarrow \infty$ is immediate from (3.9). For $\sigma_\varepsilon \rightarrow 0$: when the complete-information game has a unique equilibrium with gap $\delta_v > 0$, all thresholds are pushed into the density tails, yielding the convergence rate $\gamma(\sigma_\varepsilon) \leq (C_g/\sigma_\varepsilon) \exp(-c \delta_v w_{\min}/(2\sigma_\varepsilon)) \|J_v\|_\infty$ for log-concave g_0 with tail bound $g_0(z) \leq C_g e^{-c|z|}$ (Appendix C, Step 4). At any σ_ε where $\gamma < 1$, the implicit function theorem ensures that the BNE extends as a unique C^1 branch. Since $\gamma \rightarrow 0$ at both extremes and $\gamma_{\max} < 1$ implies $\gamma < 1$ everywhere, uniqueness holds for all $\sigma_\varepsilon > 0$. $\square$

The interaction matrix Λ controls *both* margins through a single channel. On the intensive margin, $\rho(\Lambda) < 1$ guarantees pricing uniqueness (Proposition 11). On the extensive margin, $\rho(\Lambda)$ enters the contraction modulus (3.9) through the amplification factor $1/(1 - \rho(\Lambda))$: the sensitivity of the pricing equilibrium to a demand perturbation is bounded by $\|(I - \Lambda)^{-1}\| \leq 1/(1 - \rho(\Lambda))$. The infinity-norm condition $\|\Lambda\|_\infty < 1$ of Corollary 12 is a conservative sufficient condition; the spectral condition $\rho(\Lambda) < 1$ (Proposition 11) is tighter and is used in all numerical diagnostics. At the baseline parameterization of Section 5 and Appendix F, $\rho(\Lambda) = 0.096$ (amplification factor 1.11): the outer activation problem is *nearly* a constant-markup entry game. Across the robustness experiments (Appendix F), $\rho(\Lambda)$ ranges from 0.025 (low σ , high η) to 0.574 (high σ , low η , concentrated market), remaining well below one throughout. When $\rho(\Lambda) \rightarrow 1$, the amplification diverges and larger σ_ε is required to restore the contraction.

Remark 16 (Structural vs. CES-specific elements). The Jacobian decomposition (3.8) and the bathtub shape of $\gamma(\sigma_\varepsilon)$ are *structural*: they hold for any Bayesian game with private cost shocks, a contractive pricing map, and Lipschitz-continuous marginal values—regardless of the demand system. What the CES structure provides is closed-form expressions for

the Lipschitz constants (L_D, L_μ, L_C) , the feedback coefficient κ_{nf} , and the share thresholds $\bar{s}^*$ and $\bar{s}$. Under more general demand systems (e.g., translog), markups need not be bounded, so the Lipschitz constants may diverge and the contraction modulus may exceed one. The qualitative insight—that uniqueness is governed by a bathtub-shaped modulus driven by three interpretable forces—extends beyond CES, but the quantitative verification $\gamma_{\max} < 1$ would require numerical Jacobian bounds tailored to the specific demand system.

Remark 17 (Role of private information). Proposition 15 establishes uniqueness of the Bayesian game for a given noise distribution G and scale σ_ε . The BNE is a well-defined equilibrium concept in its own right—not merely a device for selecting among complete-information equilibria. Non-marginal sites ($|\Delta\pi_{if}|$ bounded away from zero) activate with probability near one or zero; only marginal sites ($|\Delta\pi_{if}| \approx 0$) have interior activation probabilities that depend on G . When the complete-information game has a unique Nash equilibrium (as at all tested parameterizations), the BNE converges to it as $\sigma_\varepsilon \rightarrow 0$ regardless of G .

The following theorem consolidates all assumptions for the main result in one place, referencing Propositions 6–15 as building blocks. Conditions (a)–(b) are structural properties of the model; condition (c) is a distributional regularity; conditions (d)–(e) are verifiable from model primitives without solving the game—condition (d) by running the Jia–Tarski algorithm from both the supremum and infimum starting points (a zero gap certifies uniqueness), and condition (e) by evaluating γ on a grid of σ_ε values.

Theorem 18 (Main Result: Global BNE Uniqueness). *Suppose:*

- (a) $(\sigma - 1)/\theta > 1$ and each firm’s activation profit is supermodular (Proposition 6, certified analytically or computationally);
- (b) $\rho(\Lambda(v)) < 1$ for every activation profile $v \in [0, 1]^M$ in the threshold map’s image, where $\Lambda(v)$ is the markup interaction matrix at activation profile v (Proposition 11). This hypothesis admits three operational tests: (i) Corollary 19 gives a closed-form sufficient condition $\max_{n,f} s_{nf} < \bar{s}$ in terms of the maximum market share; (ii) because Λ is block-diagonal across markets, the global spectral radius satisfies $\rho(\Lambda) = \max_n \rho(\Lambda_n)$, so condition (b) reduces to per-market pricing uniformity, which admits the direct enumeration certificate in Proposition 21 below; (iii) when neither (i) nor (ii) is available, the continuity of $\rho(\Lambda(v))$

on $[0, 1]^M$ permits a finite-grid evaluation with sufficient resolution to bound the variation of ρ between grid points.¹

- (c) the standardized distribution G_0 has log-concave tails (e.g., normal or logistic);
- (d) the complete-information game has a unique Nash equilibrium with gap $\delta_v = \min_{i,f} |\Delta\pi_{if}^*|/w_i > 0$ (verifiable computationally: the Jia–Tarski algorithm from supremum and infimum starting points converges to the same profile);
- (e) $\gamma_{\max} \equiv \max_{\sigma_\varepsilon > 0} \gamma(\sigma_\varepsilon) < 1$ (verifiable by evaluating γ on a grid of σ_ε values).

Then the Bayesian Nash equilibrium is unique for all $\sigma_\varepsilon > 0$.

Proof. Conditions (a)–(b) ensure that the single-agent CDCP is supermodular and the pricing equilibrium is unique for every activation profile, so the threshold map T in (3.7) is well-defined on $[0, 1]^M$. Conditions (c)–(d) ensure that $\gamma(\sigma_\varepsilon) \rightarrow 0$ as $\sigma_\varepsilon \rightarrow 0$: log-concave tails provide the density decay, and the strict gap $\delta_v > 0$ pushes all thresholds into the tails. The limit $\gamma \rightarrow 0$ as $\sigma_\varepsilon \rightarrow \infty$ follows from $\|g_0\|_\infty/\sigma_\varepsilon \rightarrow 0$. Condition (e) ensures the peak contraction modulus is below one, so T is a contraction in the sup-norm for all $\sigma_\varepsilon > 0$. Banach’s fixed-point theorem on the complete metric space $[0, 1]^M$ yields a unique fixed point. $\square$

Sharp market-share threshold. Theorem 18 consolidates the assumptions; the following corollary provides a closed-form diagnostic. Since the feedback coefficient κ_{nf} depends only on the market share s_{nf} and the elasticity pair (σ, η) , the infinity-norm condition of Corollary 12 reduces to a threshold on the maximum market share.

Corollary 19 (Market-Share Threshold for Pricing Contraction). *Define $\bar{s}(\sigma, \eta, F)$ as the unique solution in $(0, 1)$ of $\kappa(\bar{s}; \sigma, \eta) = 1/(F - 1)$. If $\max_{n,f} s_{nf} < \bar{s}$, then the pricing equilibrium is unique (condition (b) of Theorem 18 is satisfied). Combined with conditions (a) and (c)–(e), this yields BNE uniqueness. For $(\sigma, \eta) = (5, 2)$, the threshold solves*

$$24\bar{s}^2 - 9\bar{s} - 8 = 0, \quad \bar{s} \approx 0.795 \quad (F = 2). \quad (3.10)$$

¹The infinity-norm sufficient condition $\max \kappa < 1/(F - 1)$ (Corollary 12) is uniform by construction but conservative. The spectral condition $\rho(\Lambda) < 1$ is tighter, and the enumeration certificate verifies it directly on the entire activation lattice. At all tested parameterizations in Section 5, $\rho(\Lambda) \leq 0.316$, with enumeration margin 0.68 below one.

For $F = 8$ firms, $\bar{s} \approx 0.339$; for $F = 5$ (the cloud application of Section 5), $\bar{s} \approx 0.448$; for $F = 4$, $\bar{s} \approx 0.513$.

Proof. From (3.2), for $(\sigma, \eta) = (5, 2)$: $1/\epsilon = (1 - s)/5 + s/2 = (2 + 3s)/10$, so $\mu = 10/(8 - 3s)$ and $\beta = 12/(8 - 3s)$. Substituting into κ :

$$\kappa(s) = \frac{12s^2}{8 + 9s - 12s^2}.$$

Setting $\kappa = 1$ (i.e., $F = 2$) gives $24s^2 - 9s - 8 = 0$, whose positive root is $\bar{s} = (9 + \sqrt{849})/48 \approx 0.795$. $\square$

Comparative statics follow from κ being increasing in s and in $\sigma - \eta$:

1. $\partial \bar{s} / \partial \sigma < 0$: higher within-sector substitution (larger $\sigma - \eta$) strengthens the markup channel, *tightening* the threshold.
2. $\partial \bar{s} / \partial \eta > 0$ (for $\eta < \sigma$): as $\eta \rightarrow \sigma$, nested CES collapses to standard CES with constant markups, so $\kappa \rightarrow 0$ and $\bar{s} \rightarrow 1$.

Table 2 reports $\bar{s}$ for selected (σ, η, F) combinations, showing that the threshold is tightest when the markup channel is strongest (low η , high $\sigma - \eta$, large F).

Remark 20 (Two share thresholds). The paper employs two distinct market-share thresholds, both *sufficient* conditions that are conservative. The *supermodularity threshold* $\bar{s}^* \equiv (\alpha - 1)/(2\alpha)$ from Proposition 6 governs per-market convexity under the CES benchmark and depends only on (σ, θ) . The *pricing-contraction threshold* $\bar{s}(\sigma, \eta, F)$ from Corollary 19 governs pricing uniqueness via $\|\Lambda\|_\infty$ and additionally depends on (η, F) . At the baseline $(\sigma, \eta, \theta) = (5, 2, 3)$ with $F = 8$: $\bar{s}^* = 0.125$ and $\bar{s} \approx 0.339$, while $\max_{n,f} s_{nf} = 0.39$ —both thresholds are violated. Nevertheless, aggregate supermodularity holds (Jia–Tarski gap = 0; Remark 7) and pricing uniqueness holds under the tighter spectral condition ($\rho(\Lambda) = 0.096 \ll 1$). The share thresholds are useful as closed-form diagnostics but are not necessary conditions.

TFP dispersion and marginal sites. Fix all parameters except TFP and define the dispersion ratio $\delta = a_{\max}/a_{\min}$. As δ increases, the number of marginal sites weakly decreases: in the limit $\delta \rightarrow \infty$, only the highest-TFP firms are active in any market, and $\kappa \rightarrow 0$ for

Table 2: Market-share threshold $\bar{s}(\sigma, \eta, F)$ for pricing contraction

$\sigma \backslash \eta$	$\kappa < 1$ ($F = 2$)				$\kappa < 1/4$ ($F = 5$, cloud app)				$\kappa < 1/7$ ($F = 8$)			
	1.5	2.0	3.0	4.0	1.5	2.0	3.0	4.0	1.5	2.0	3.0	4.0
3	0.843	1.000	—	—	0.500	0.684	—	—	0.386	0.532	—	—
5	0.687	0.795	1.000	1.000	0.377	0.448	0.628	0.973	0.283	0.339	0.482	0.756
8	0.616	0.675	0.802	0.953	0.317	0.358	0.443	0.541	0.232	0.265	0.333	0.411
10	0.593	0.640	0.736	0.841	0.297	0.330	0.395	0.465	0.214	0.241	0.294	0.350

non-dominant firms. When $a_f = a$ for all f , market shares are balanced at approximately $s \approx 1/F$. Since κ_{nf} is increasing and convex in s_{nf} , balanced shares actually *minimize* the feedback coefficient: the pricing contraction condition (Proposition 11) is *easier* to satisfy under identical TFP. Multiplicity in this case requires marginal sites driven by the available-set structure $\{\mathcal{A}_f\}$, which is absent at the baseline. Appendix F confirms this prediction numerically.

Direct enumeration certificate. The block-diagonal structure of Λ across markets permits a stronger verification of condition (b) than finite-grid evaluation: because each block Λ_n depends only on market- n 's activation and on calibrated marginal costs, one can enumerate over all $2^F - 1$ nonempty per-market subsets and certify uniformity directly. The following proposition records the result.

Proposition 21 (Direct Enumeration Certificate). *Let Λ be block-diagonal with blocks $\Lambda_n(S_n)$, where $S_n \subseteq \{1, \dots, F\}$ is the set of firms active in market n . Then $\rho(\Lambda) = \max_n \rho(\Lambda_n(S_n))$, and condition (b) of Theorem 18 holds whenever*

$$\bar{\rho} \equiv \max_n \max_{\emptyset \neq S \subseteq \{1, \dots, F\}} \rho(\Lambda_n(S)) < 1,$$

where the inner maximum is over the finite set of per-market activation subsets and the per-market subgame is solved at the calibrated marginal-cost vector. The enumeration comprises $N(2^F - 1)$ pricing subgames and is exhaustive on the activation lattice $\{0, 1\}^{NF}$.

Proof. Block-diagonality of Λ follows from Proposition 11: the interaction $\Lambda_{n,fg}$ is nonzero only when $n = n'$, so the spectrum of Λ is the union of per-block spectra and $\rho(\Lambda) = \max_n \rho(\Lambda_n)$. Each Λ_n depends on market n only through the activation subset S_n and the equilibrium shares $(s_{n,f})_{f \in S_n}$, which are uniquely determined by the per-market Bertrand fixed point given the calibrated marginal costs $(c_{n,f})_{f \in S_n}$ (by Proposition 11 once $\bar{\rho} < 1$ is

Table 3: Direct Enumeration Certificate for Theorem 18(b)

Statistic	Value
Subgames solved	1302
Markets $\times$ subsets	42×31
$\max_{n,S} \rho(\Lambda_n(S))$	0.3158
99th percentile	0.3144
95th percentile	0.3097
Median	0.1663
Mean	0.1711
Margin below 1	0.6842
Argmax market	Hong Kong
Argmax $ S $	2 firms

Notes: Per-market Bertrand subgames solved for each nonempty subset $S \subseteq \{1, \dots, 5\}$ across all 42 markets. $\rho(\Lambda_n(S))$ computed at the calibrated marginal-cost vector. Base-line parameters ($\sigma = 5$, $\eta = 2$, Spec A). Because Λ is block-diagonal across markets, $\max_{n,S} \rho(\Lambda_n(S)) < 1$ certifies pricing uniqueness on the entire activation lattice $\{0, 1\}^{NF}$.

verified, and by a finite sweep otherwise). The activation lattice $\{0, 1\}^{NF}$ decomposes as $\prod_n \{0, 1\}^F$, so enumerating per-market subsets covers it exhaustively. $\square \quad \square$

For the calibrated cloud economy ($N = 42$, $F = 5$), enumeration over all $42 \times 31 = 1,302$ per-market subgames yields $\bar{\rho} = 0.316$ (Table 3), giving a margin of 0.684 below one. The maximum is attained in a two-firm {Oracle, Alibaba} subgame in Hong Kong, where one firm’s price is so much lower than the other’s that it takes almost the entire subgame market. Even in such corner configurations, the spectral radius is an order of magnitude below one: the infinity-norm bound $\max \kappa < 1/(F - 1)$ is violated in $\sim 12\%$ of the enumerated subgames (a CES-ratio peak of 5.1), but $\rho(\Lambda_n)$ remains bounded by 0.32 because the rows of Λ_n concentrate on rival firms whose own κ ’s are simultaneously small. The spectral condition is therefore substantially tighter than Corollary 19, and the enumeration certificate verifies it globally.

3.3 The Bathtub Modulus: Private-Cost Noise as Uniqueness Discipline

The decomposition (3.8) yields the paper’s central economic insight, which we separate here because it is conceptually distinct from the mechanical uniqueness proof. The con-

traction modulus $\gamma(\sigma_\varepsilon)$ has a *bathtub* shape in the noise scale σ_ε : uniqueness tightens at both ends of the private-information spectrum, and loosens only at intermediate scales where entry decisions are most coupled.

Two vanishing tails. The two tails dissipate multiplicity through different channels:

- *Large σ_ε (diffuse beliefs).* The density prefactor $g_0(\cdot)/\sigma_\varepsilon$ vanishes: each marginal shift in rivals' activation probabilities is smoothed across a wide distribution of thresholds, so the best-response map becomes nearly flat. Noise itself is the discipline.
- *Small σ_ε (near-complete information).* When the complete-information equilibrium is unique with strictly positive marginal-value gap δ_v , every threshold lies in the tail of g_0 , and the density evaluation at $g_0(\bar{\varepsilon}/\sigma_\varepsilon)$ decays exponentially. Sites become far-from-marginal and their activation probabilities are locally insensitive to perturbations of rivals' strategies.

Multiplicity therefore concentrates at intermediate σ_ε where some thresholds sit near the density mode. The peak $\gamma_{\max}$ is the single number that witnesses global uniqueness: if $\gamma_{\max} < 1$, the BNE is unique for every $\sigma_\varepsilon > 0$ (Proposition 15(iv)).

Verification strategy. The condition $\gamma_{\max} < 1$ is verifiable from primitives without enumerating equilibria: evaluate $\gamma(\sigma_\varepsilon)$ on a log-spaced grid covering $(0, \infty)$ and take the maximum. Because γ is continuous in σ_ε (implicit function theorem at every interior point) and vanishes at both extremes, a finite grid with sufficient margin below one certifies uniqueness globally. Figure 10 illustrates the bathtub in a small economy ($N = 3$, $F = 4$) where the Jacobian can be enumerated exactly; the peak $\gamma_{\max} = 0.951 < 1$ lies at $\sigma_\varepsilon/\bar{\phi} \approx 0.3$. The same numerical procedure applied to the calibrated cloud economy (Section 5) yields $\gamma_{\max} \ll 1$ at all parameterizations (Appendix F).

Why this is new. Standard uniqueness results in entry games deliver global selection only when strategies are binary (Espin-Sanchez et al. 2023), payoffs are strictly monotone (Harrison and Jara-Moroni 2021), or equilibrium is approximated by a many-firm limit (Weintraub et al. 2008). The bathtub modulus instead exploits the structural fact that private-cost noise enters the Jacobian multiplicatively through $g_0(\cdot)/\sigma_\varepsilon$: intermediate noise can fail contraction only over a compact window, and the two tails vanish by fundamentally

different economic mechanisms. The criterion $\gamma_{\max} < 1$ is a single primitive-level check, independent of the number of firms, the size of the action space, or the equilibrium-selection rule.

3.4 Example: Dimensionality Reduction through Combinatorial Structure

We illustrate how the CDCP approach reduces dimensionality relative to the binary-entry framework of [Espin-Sanchez et al. \(2023\)](#) using a small economy (hereafter the *small calibration*): $F = 4$ firms, $N = 3$ markets, with two “strong” firms ($a_f = 1.5, 1.2$; access to all 3 sites) and two “weak” firms ($a_f = 1.0, 0.8$; access to 2 sites each). The complete-information equilibrium is unique (supremum and infimum Jia–Tarski iterates coincide) with all firms active at their full site sets.

The [Espin-Sanchez et al.](#) approach. Treating each firm-site pair (i, f) as an independent binary player gives $M = 10$ players in a threshold system $v_{if} = G_0(\bar{\varepsilon}_{if}/\sigma_\varepsilon)$. The configuration space has $2^{10} = 1,024$ joint action profiles. Each weak firm has one clearly profitable site (marginal value 0.28–0.71 from empty) and one marginal or unprofitable site (MV ≈ -0.003 to -0.14 from empty). Under [Espin-Sanchez et al.](#), these marginal individual-site decisions are independent threshold variables that expand the fixed-point dimension.

The CDCP approach. Recognizing that each firm solves a combinatorial problem over $2^{|\mathcal{A}_f|}$ configurations collapses the 10-player game into a 4-player game. Within-firm complementarity bundles the weak firms’ marginal sites with their profitable sites: for both weak firms, the Fréchet cost complementarity creates positive increasing differences (ID = 0.020 and 0.044), so that each site is profitable conditional on the other even when individually marginal. The CDCP optimal set is the full site set $\mathcal{A}_f$ for each firm, and the 4-player threshold system has 60% fewer degrees of freedom.

Takeaway. The combinatorial structure collapses $M = \sum_f |\mathcal{A}_f|$ binary decisions into F firm-level decisions—a 60% reduction in threshold dimension for this example. The reduction is economically, not merely computationally, substantive: within-firm cost

complementarity ($ID > 0$ for both weak firms) bundles individually marginal sites into determinate firm-level packages. Viewing each site as an independent binary player (Espin-Sanchez et al.) therefore *over-states* the effective dimension of the threshold system; the CDCP approach identifies which sites are separable decisions and which are bundled with complementary sites, before checking uniqueness. Two consequences follow for the downstream analysis. First, the lower-dimensional threshold system yields a tighter row-sum Jacobian bound in the contraction analysis of Section 3.2, because within-firm complementarities are internalized rather than treated as additional cross-player interactions; this directly lowers $\gamma_{\max}$ and widens the parameter region where the bathtub criterion delivers uniqueness. Second, in configurations where within-firm ID is large enough that the CDCP optimal set *strictly exceeds* the union of sites individually above threshold, the binary-action framework cannot certify uniqueness without ad hoc coordination assumptions, while the CDCP structure pins down the optimal set by lattice arguments. The 4-firm small economy has uniqueness under both formulations, but the benefit of the combinatorial framing scales with the strength of within-firm complementarity. Numerical details are in the companion script `main_example_2x2.m`.

4. COMPUTATIONAL ALGORITHM

The equilibrium defined in Section 2 is computed by a three-layer nested algorithm. We describe each layer and present pseudocode; Appendix E reports performance diagnostics for four algorithm variants and a 40-country extension demonstrating scalability. Figure 2 illustrates the structure.

4.1 Three-Layer Structure

Layer 1 (Innermost): Atkeson–Burstein markup fixed point. For given marginal cost c_{nf} and rival demand index $D_{-f}(n)$, the equilibrium markup μ_{nf} is the fixed point of:

$$\mu_{nf}^{(t+1)} = \frac{\epsilon_{nf}^{(t)}}{\epsilon_{nf}^{(t)} - 1}, \quad \frac{1}{\epsilon_{nf}^{(t)}} = \frac{1 - s_{nf}^{(t)}}{\sigma} + \frac{s_{nf}^{(t)}}{\eta}, \quad (4.1)$$

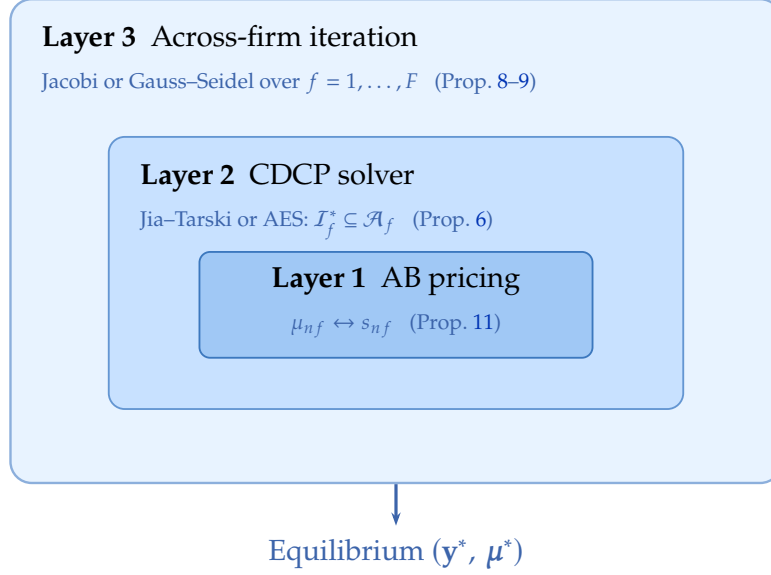


Figure 2: Three-layer algorithm nesting. Each profit evaluation at Layer 2 calls the AB pricing fixed point (Layer 1); the across-firm iteration at Layer 3 updates rival demand indices until convergence. Four algorithm variants arise from the 2×2 choice of solver at Layers 2 and 3 (Appendix E).

where $s_{nf}^{(t)} = (\mu_{nf}^{(t)} c_{nf})^{1-\sigma} / [(\mu_{nf}^{(t)} c_{nf})^{1-\sigma} + D_{-f}(n)]$. The iteration is initialized at the CES markup $\mu^{(0)} = \sigma/(\sigma - 1)$ and damped with weight $\omega_\mu = 0.5$:

$$\mu^{(t+1)} \leftarrow (1 - \omega_\mu) \mu^{(t)} + \omega_\mu \mu^{(t+1)}.$$

Convergence is guaranteed by Proposition 11 whenever $\rho(\Lambda) < 1$.

Layer 2 (Middle): Single-agent CDCP. Given the rival demand index $D_{-f}(n)$ for all markets n , each firm f solves the combinatorial optimization:

$$\mathcal{I}_f^* \in \arg \max_{\mathcal{I}_f \subseteq \mathcal{A}_f} \pi_f(\mathcal{I}_f; D_{-f}), \quad (4.2)$$

where each profit evaluation calls the AB markup fixed point (Layer 1). Two inner solvers can be used:

- (a) **Jia-Tarski bounds** (Jia, 2008): Compute an *upper bound* $\overline{\mathcal{I}}_f$ by starting from all candidates $\mathcal{A}_f$ and iteratively removing elements with negative marginal value; compute a *lower bound* $\underline{\mathcal{I}}_f$ by starting from $\emptyset$ and iteratively adding elements with

positive marginal value. If $\bar{\mathcal{I}}_f = \underline{\mathcal{I}}_f$, the solution is exact.

- (b) **AES branch-and-bound** (Arkoulakis et al., 2023): Alternate between a *squeezing phase* (simultaneously squeeze included/excluded sets by evaluating marginal values against both the lower and upper bounds of the candidate set) and a *branching phase* (recursively split on an undecided candidate). The squeezing phase typically resolves most candidates, leaving few branches.

Both solvers find the exact optimum. When the equilibrium shares satisfy the conditions of Proposition 6, the Jia–Tarski bounds converge without residual gap, computationally certifying that supermodularity holds at the equilibrium.

Layer 3 (Outermost): Cross-firm iteration. The rival demand index $D_{-f}(n) = \sum_{f' \neq f} (\mu_{nf'} c_{nf'})^{1-\sigma}$ depends on all firms' equilibrium prices, which in turn depend on their activation sets. Two outer iteration methods resolve this fixed point:

- (a) **Jacobi** (simultaneous best response): In each round k , every firm f computes its best-response activation set $\mathcal{I}_f^{(k+1)}$ against the rival demand $D_{-f}^{(k)}$ from the *previous* round. All firms update simultaneously, and price-power contributions are damped:

$$p_{nf}^{(k+1)} \leftarrow (1 - \omega) p_{nf}^{(k)} + \omega \tilde{p}_{nf}^{(k+1)},$$

where $\tilde{p}_{nf}^{(k+1)}$ is the price implied by $\mathcal{I}_f^{(k+1)}$ and $\omega \in (0, 1]$ is a damping parameter. Damping does not affect the set of fixed points—only the convergence path—because any fixed point satisfies $\tilde{p} = p$.

- (b) **Gauss–Seidel** (sequential best response): Firms are processed in a fixed order $(f_1, \dots, f_F)$. When firm f_j updates, it uses the most recent price-power contributions from firms $f_1, \dots, f_{j-1}$ (already updated in this round) and the previous round's contributions from $f_{j+1}, \dots, f_F$.

By Proposition 9, Jacobi iteration converges monotonically to extremal equilibria when initialized from the supremum or infimum; Gauss–Seidel convergence is empirically robust (Remark 10).

Algorithmic recommendation. Propositions 6–9 yield a concrete recipe: initialize the outer iteration from the *supremum* (all available sites active) and solve each firm’s CDCP via the Jia–Tarski or AES algorithm. Starting from the supremum guarantees convergence to the upper bound of the equilibrium set (Proposition 9), which at all tested parameterizations is itself the unique Nash equilibrium. The nested CES structure further accelerates computation: the interaction matrix Λ is block-diagonal across markets (N blocks of size $F \times F$), so $\rho(\Lambda) = \max_n \rho(\Lambda_n)$ is computable in $O(NF^3)$, and the closed-form feedback coefficient (3.2) reduces the contraction check to a single market-share comparison.

4.2 Pseudocode

Algorithm 1 summarizes the full procedure.

Algorithm 1 Three-Layer Equilibrium Solver

Input: Parameters $(\sigma, \eta, \theta, \{a_f\}, \{\phi_i, w_i, \tau_{in}\}, \{\Omega_n\})$; inner solver $\in \{\text{JT}, \text{AES}\}$; outer method $\in \{\text{Jacobi}, \text{GS}\}$

Output: Equilibrium $(\{\mathcal{I}_f^*\}, \{\mu_{nf}^*\}, \{s_{nf}^*\})$

```

1: Initialize:  $\mathcal{I}_f^{(0)} \leftarrow \mathcal{A}_f$  for all  $f$ ; compute  $p_{nf}^{(0)}$  at CES markups
2: for  $k = 0, 1, 2, \dots$  do
3:   for  $f = 1, \dots, F$  do ▷ Jacobi: use  $D^{(k)}$ ; GS: use latest  $D$ 
4:      $D_{-f}(n) \leftarrow \sum_{f' \neq f} [p_{nf'}^{1-\sigma}]^{\text{current}}$  for all  $n$ 
5:      $\mathcal{I}_f^{(k+1)} \leftarrow \text{INNER SOLVER}(f, D_{-f}, \text{par})$  ▷ Layer 2: CDCP
6:     Update  $p_{nf}$  via  $\text{ABMARKUP}(c_{nf}(\mathcal{I}_f^{(k+1)}), D_{-f})$  ▷ Layer 1
7:   end for
8:   if  $\max_{n,f} |p_{nf}^{(k+1)} - p_{nf}^{(k)}| < \text{tol}$  then break
9:   end if
10: end for

```

5. QUANTITATIVE ILLUSTRATION: CLOUD INFRASTRUCTURE ENTRY

We apply the framework to the global cloud infrastructure industry, where five dominant providers—Amazon Web Services (AWS), Microsoft Azure (MSFT), Google Cloud (GOOG), Oracle Cloud Infrastructure (ORCL), and Alibaba Cloud (BABA)—choose which countries to establish cloud regions. This industry provides a natural setting for the model: a small number of heterogeneous firms ($F = 5$) making discrete entry decisions across a

large number of potential markets ($N = 42$ countries), with high fixed costs, observable activation patterns, and variable markups that respond to local competitive conditions.

The calibration follows the identification structure of [Arkolakis et al. \(2023\)](#)—fixing demand-side objects and firm scale externally, then calibrating fixed-cost parameters from entry moments—though with a thinner moment basis: binary entry counts rather than the affiliate counts, survival rates, and gravity coefficients available in AES. We fix firm productivity a_f from revenue data and market demand Ω_n from GDP, then calibrate market-level entry costs ϕ_i and a firm-specific deployment-efficiency wedge g_f to match the observed provider-country activation matrix. This strategy separates the scale question (handled by revenue data) from the geographic-presence question (handled by the model). We evaluate three calibration specifications that differ in which parameters are taken from data versus calibrated; the preferred specification achieves 99.5% agreement with the observed activation matrix. All equilibrium computations use the solver described in [Section 4](#), unchanged.

5.1 Data and Sample

The unit of observation is a *cloud region*: a named availability zone where a provider offers compute, storage, and networking services from local infrastructure. A cloud region implies at least one datacenter facility and a commitment to serve customers with local latency guarantees. We source region lists from each provider’s official documentation (Q1 2026 vintage) and Datacenter Map (purchased database, April 9, 2026 vintage; <https://www.datacentermap.com>).

Table 4 summarizes the sample. The five firms operate a combined 129 cloud regions across 42 countries, with feasible sets ranging from 14 (Alibaba) to 34 (Microsoft). AWS has 5 additional pipeline regions (announced, under construction), which enter only the feasible set $\mathcal{A}_f$, not the observed activation $\mathcal{I}_f^{\text{data}}$.

Why five firms. AWS, Azure, and Google Cloud account for roughly 65% of global cloud infrastructure revenue. Adding Oracle and Alibaba brings coverage to approximately 80% and, crucially, introduces asymmetry absent in a Big 3 specification: Alibaba’s presence is concentrated in East and Southeast Asia, while Oracle pursues an aggressive multi-region expansion into emerging markets. The resulting feasible-set heterogeneity ($|\mathcal{A}_f|$ ranges

Table 4: Data and Sample Construction

Firm	Provider	Operational regions	Pipeline regions	Feasible markets
AWS	Amazon Web Services	25	5	30
MSFT	Microsoft Azure	34	0	34
GOOG	Google Cloud	29	0	29
ORCL	Oracle Cloud	27	0	27
BABA	Alibaba Cloud	14	0	14
Total		129		

Notes: Sample covers 42 countries/territories. Operational regions: firm has at least one cloud-infrastructure facility with local compute. Pipeline: announced or under construction. Feasible markets: operational + pipeline (revealed lower bound on $\mathcal{A}_f$). Source: official provider region pages and Datacenter Map (purchased database, April 9, 2026 vintage; <https://www.datacentermap.com>).

from 14 for Alibaba to 34 for Microsoft) generates richer extensive-margin counterfactuals.

5.2 Calibration

The calibration follows the identification structure of [Arkolakis et al. \(2023\)](#): demand-side objects and firm scale are set externally, while fixed-cost parameters are calibrated from entry moments. Full details are in Appendix D.

Externally set parameters. Firm TFP a_f is fixed using cloud segment revenue, mapped log-linearly into $[1.0, 2.0]$: AWS ($a_f = 2.00$), Microsoft (1.76), Google (1.48), Oracle (1.09), Alibaba (1.00). Market demand is $\Omega_n = (\text{GDP}_n / \text{GDP}_{\text{US}})^{0.7}$, where the concavity reflects that cloud spending scales less than proportionally with GDP; $\beta = 0.7$ is chosen from a parsimonious calibration comparison rather than estimated from independent demand data (Appendix D). Local costs w_i are PWT human-capital-adjusted compensation per worker. Trade costs follow a discrete three-tier structure: $\tau = 1.0$ (same country), 1.15 (same continent), 1.5 (cross-continent). We retain $\sigma = 5$, $\eta = 2$, $\theta = 3$, giving $(\sigma - 1)/\theta = 4/3 > 1$.

Calibrated parameters. Market-level entry costs ϕ_i and firm-specific deployment-efficiency wedges g_f are calibrated from the observed activation matrix. The fixed-cost structure is

$$\text{fixed cost}_{fi} = \phi_i \cdot g_f \cdot w_i, \quad (5.1)$$

where ϕ_i captures non-wage market costs (regulation, infrastructure) and g_f captures firm-specific deployment efficiency ($g_{\text{AWS}} = 1$ by normalization). The calibration targets per-market entry counts (identifying ϕ_i) and per-firm totals (identifying g_f) via alternating coordinate descent.

Identification and limitations. Unlike [Arkolakis et al. \(2023\)](#), we do not observe provider-country sales, affiliate-count analogues, or survival moments—the calibration has a thinner identification basis, relying on a single class of moments (entry counts). The goal is to illustrate the quantitative relevance of the endogenous-pricing mechanism rather than to disentangle all sources of cost heterogeneity. The counterfactual exercises focus on *qualitative patterns*—which firms are most affected, whether effects are regressive—that are robust to the specific calibrated values.

5.3 Three Calibration Specifications

We evaluate three calibration specifications that differ in which firm-level parameters are set externally versus calibrated from entry moments. The preferred specification (Spec A) fixes a_f at revenue-based values and calibrates (g_f, ϕ_i) ; two alternatives calibrate a_f from moments with or without g_f (Appendix D). Spec A matches 209 of 210 firm-market cells in sample (99.5%), with 100% precision and 99.2% recall. It dominates the alternatives despite having the same number of free firm parameters, because fixing a_f externally separates firm scale from deployment efficiency. The single mismatch occurs at a market on the entry-indifference margin; the model generates no false positives.

Is g_f doing real work? To isolate the role of the deployment-efficiency wedge, we recalibrate with $g_f \equiv 1$ (Spec D) and compare to the preferred Spec A. Table 5 summarizes the contrast. With g_f switched off, the calibration must absorb all cross-firm variation into a purely market-level ϕ_i ; fit drops from 99.5% to 87.1% cell agreement (42 exact-match markets fall to 18), and precision/recall fall to 96.4% / 82.2%. Crucially, however, the AB-vs-CES comparison is essentially invariant to the specification: the profit decline is -30.1% under Spec A and -33.3% under Spec D, with an identical -15 active-pair change. The cross-specification activation correlation is 0.74. In short, g_f matters for *levels* of fit but not for the qualitative mechanism: the intensive-margin AB channel survives imposition

Table 5: Role of deployment-efficiency wedge g_f : Spec A vs Spec D

	Spec A ($\phi_i \cdot g_f$)	Spec D ($\phi_i, g_f \equiv 1$)
<i>Uniqueness & interaction</i>		
Active pairs (supremum)	128	110
Jia–Tarski gap	0	0
$\rho(\Lambda)$	0.103	0.123
$\max_{n,f} \kappa_{nf}$	0.193	0.230
<i>In-sample fit</i>		
Cell agreement (%)	99.5	87.1
Precision (%)	100.0	96.4
Recall (%)	99.2	82.2
Per-market entry MAE	0.02	0.60
Exact-match markets / 42	41	18
<i>AB $\rightarrow$ CES counterfactual</i>		
Profit change (%)	-30.1	-33.3
Active-pair change	-15	-15
<i>Cross-spec consistency</i>		
$\text{corr}(I_A, I_D)$	0.742	

Notes: Spec A calibrates (g_f, ϕ_i) jointly; Spec D imposes $g_f \equiv 1$ and calibrates ϕ_i only. Fit statistics computed against the 210-cell observed activation matrix. AB $\rightarrow$ CES columns report the change in aggregate profit and active-pair count when variable markups are replaced by the CES constant $\mu = \sigma/(\sigma - 1)$.

of $g_f \equiv 1$.

5.4 Calibrated Parameters

Table 6 reports the Spec A calibrated parameters.

The deployment-efficiency wedges g_f have a clear pattern: Microsoft and Google have similar efficiency ($g_f \approx 0.49$ and 0.42), while Oracle and Alibaba require much lower g_f (0.15 and 0.12) to rationalize their broad entry despite modest revenue-based scale. We emphasize that g_f is a reduced-form wedge, not a literal engineering parameter; it does not vary across counterfactuals. The 42 calibrated ϕ_i span three orders of magnitude, from 10^{-4} (UAE) to 0.26 (Chile); only one market sits at the calibration lower bound.

Hub plausibility. The calibrated fixed-cost distribution is economically plausible: the three regional hyperscaler hubs rank in the bottom half of ϕ_i percentiles, consistent with these markets serving as low-friction entry points for global cloud infrastructure

Table 6: Calibrated Parameters: Preferred Specification (Spec A)

Panel A: Firm parameters				
Firm	a_f	g_f	Markets	Interpretation
AWS	2.00	1.00	24	Numeraire; largest by revenue
MSFT	1.76	0.49	34	Broadest geographic presence
GOOG	1.48	0.42	29	Strong in developed markets
ORCL	1.09	0.15	27	Low g_f : broad despite low scale
BABA	1.00	0.12	14	Lowest g_f : niche but widespread feasible set

Panel B: Market entry costs (ϕ_i)						
	Min	p10	Median	p90	Max	p90/p10
ϕ_i	0.0001	0.001	0.002	0.044	0.262	43.6

Notes: a_f : revenue-based TFP. g_f : deployment-efficiency wedge ($g_{AWS} = 1$). ϕ_i : market entry cost. Demand: $\Omega_n = (GDP_n/GDP_{US})^{0.7}$; wages: HC-adjusted.

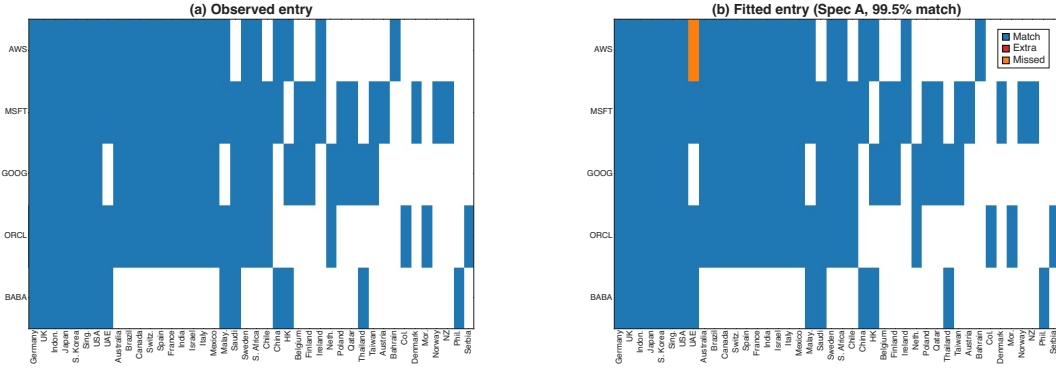


Figure 3: Observed versus fitted activation matrix (Spec A). Rows: firms (sorted by revenue-based a_f). Columns: 42 markets (sorted by calibrated ϕ_i). Dark cells: active in both data and model. Light cells with markers: mismatches (1 of 210 cells, 99.5% agreement).

deployment. Singapore is at the 31st percentile ($\phi = 0.001$), the United States at the 36th percentile ($\phi = 0.001$), and Ireland at the 43rd percentile ($\phi = 0.001$); the highest ϕ_i markets are Thailand, the Philippines, and India, whose activation requires large revenue-based variable profits to offset entry costs.

Figure 3 displays the observed versus fitted activation matrix.

5.5 Equilibrium Diagnostics

Equilibrium uniqueness holds across all specifications and counterfactuals examined. For the preferred Spec A, $\rho(\Lambda) = 0.103$ —well below the theoretical threshold of one

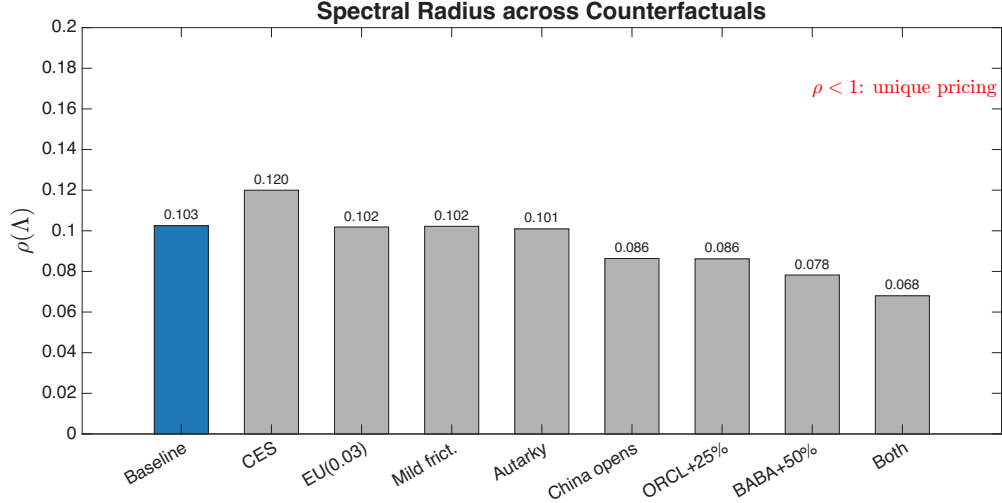


Figure 4: Spectral radius $\rho(\Lambda)$ across all counterfactual scenarios. The baseline and all counterfactuals satisfy $\rho(\Lambda) < 0.15 \ll 1$, confirming pricing uniqueness throughout.

(Proposition 11)—and the Jia–Tarski algorithm converges to the same equilibrium from both the supremum and infimum initializations (JT gap = 0). The spectral radius is stable across specifications: it ranges from 0.070 to 0.147 across an 11-point (σ, η, θ) sensitivity grid around the cloud baseline, with uniqueness holding even at $(\sigma, \theta) = (4, 4)$ where the sufficient condition $(\sigma - 1)/\theta > 1$ fails. The synthetic economy (Appendix F, Table 15) extends the range to $\rho(\Lambda) = 0.225$ at $(\sigma, \eta) = (8, 1.5)$, still well below one. Figure 4 plots $\rho(\Lambda)$ across all counterfactual scenarios: the spectral radius ranges from 0.068 to 0.120, with the JT gap equal to zero throughout. With only $F = 5$ firms and moderate market concentration, strategic complementarities are too weak to generate multiplicity at any tested parameterization. Aggregate supermodularity is supported not only by JT gap coincidence, but also by a post-hoc interval audit of the squeezing procedure: all 129 decisive inclusion/exclusion inequalities are interval-certified with zero inconclusive cases, conditional on the converged rival-demand vectors (Appendix D). Appendix F provides additional stress tests using synthetic data, including a broad (δ, α) sweep that maps the uniqueness boundary and confirms that AB markups *expand* the uniqueness region relative to CES.

Table 7: Counterfactual Results (Spec A Baseline)

Scenario	Active	Δ	$\rho(\Lambda)$	$\Delta\pi$ (%)	Most affected
Baseline	128	—	0.103	—	
<i>Markup channel</i>					
CES markup	113	−15	0.120	−30.1	ORCL, BABA
<i>EU compliance cost</i>					
$\Delta\phi=0.001$	111	−17	0.103	—	ORCL exits EU entirely
<i>China data-localization</i>					
Full autarky	128	0	0.101	−0.6	Informative null

Notes: Baseline uses Spec A (revenue-based a_f , calibrated g_f and ϕ_i ; demand $\Omega_n = (\text{GDP}_n/\text{GDP}_{\text{US}})^{0.7}$; HC-adjusted wages). Equilibrium is unique (JT gap = 0) in every scenario. CES replaces AB variable markups with constant $\mu = \sigma/(\sigma - 1)$. EU compliance adds $\Delta\phi$ to fixed costs in 12 EU markets. China autarky removes China from all non-BABA feasible sets.

5.6 Mechanism: AB versus CES Markups

The headline result from the cloud application is that endogenous variable markups—the core theoretical contribution of this article—materially sustain fringe-firm entry. This subsection presents the AB-versus-CES comparison as the primary mechanism test; the following subsections present policy counterfactuals. Table 7 summarizes all results. Equilibrium uniqueness is maintained in every scenario (JT gap = 0).

Replacing AB variable markups with constant CES markups ($\mu = \sigma/(\sigma - 1) = 1.25$) reduces active firm-market pairs from 128 to 113 (−12%) and aggregate profit by 30%. Figure 5 presents the results.

The cross-sectional mean AB markup is $\bar{\mu}^{AB} = 1.37$ with a range of [1.27, 1.47], versus the uniform $\mu^{CES} = 1.25$ (Figure 5a). The markup inflation compresses market shares: the maximum share under AB is 0.40, compared with a higher concentration under CES (Figure 5b, where all points lie below the 45-degree line). This compression is the intensive-margin channel through which endogenous markups generate pro-competitive effects.

The profit decline is not monotone in TFP rank: AWS loses 25%, Microsoft 29%, Google 42%, Oracle 40%, and Alibaba 37% (Figure 5c). The non-monotonicity arises from the interaction between the AB markup premium and the deployment-efficiency wedge g_f . Google and Oracle operate in many markets at moderate market shares where the AB premium—the gap between variable markups and the CES constant—is largest; losing this premium across a broad portfolio is costly. Alibaba, despite having the lowest TFP, is

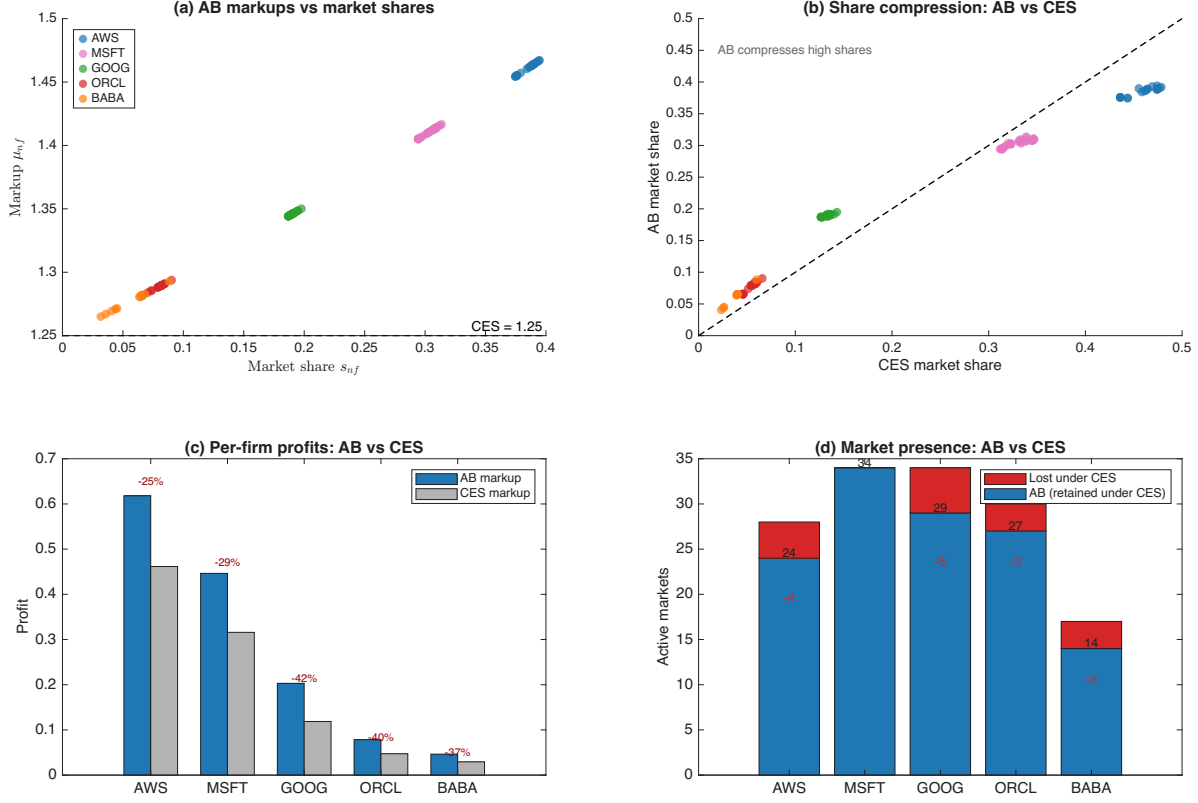


Figure 5: AB versus CES markups in the cloud application. (a) AB markup versus market share: each dot is a (firm, market) pair; dashed line marks CES constant $\mu = 1.25$. (b) Share compression: AB shares lie below the 45-degree line, indicating pro-competitive effects. (c) Per-firm profits under both pricing rules. (d) Market presence: stacked bars show markets retained vs. lost under CES.

partially insulated by its very low $g_f = 0.12$: its effective fixed costs $\phi_i \cdot g_f \cdot w_i$ are so small that even CES-level variable profits cover them in most markets. The AB mechanism thus protects firms with *non-trivial fixed costs and moderate market shares* more than firms with minimal fixed-cost exposure.

The extensive-margin effect is concentrated in Oracle and Alibaba; AWS is unaffected (Figure 5d). The result is robust across calibration specifications and demand parameterizations: the CES profit decline ranges from 30% to 31% across the preferred and robustness specifications. Under the calibrated parameters, a constant-markup CES specification understates fringe-provider participation relative to the AB benchmark.

Elasticity robustness. The headline CES profit decline depends on the demand elasticity σ , which governs both the AB markup premium and the CES benchmark $\mu^{\text{CES}} = \sigma / (\sigma - 1)$.

Table 8: σ -Robustness of the AB-versus-CES Mechanism

σ	Baseline (AB)		CES counterfactual		EU compliance ($\Delta\phi=0.03$)		
	Active	$\rho(\Lambda)$	Active	$\Delta\pi$ (%)	Active (AB)	Active (CES)	Amplification
3	116	0.028	110	-10.7	87	82	1.04
5	128	0.103	113	-30.1	97	81	0.97
6	128	0.151	107	-37.2	95	75	1.03

Notes: For each σ , all other parameters held at $\sigma = 5$ calibrated baseline (Spec A). $\rho(\Lambda)$ is the spectral radius of the markup interaction matrix at the baseline equilibrium; Active is the number of firm-market pairs in equilibrium. $\Delta\pi$ is the CES-induced change in aggregate variable profit. The rightmost three columns report the EU compliance counterfactual ($\Delta\phi = 0.03$) under AB and CES pricing; Amplification is the ratio $|\Delta n_{AB}|/|\Delta n_{CES}|$ of extensive-margin responses. Bold row is the preferred calibration ($\sigma = 5$).

We re-solve the baseline and CES counterfactual at $\sigma \in \{3, 5, 6\}$, holding ϕ_i , g_f , and a_f at their calibrated values. The CES-induced profit decline ranges from -10.7% (at $\sigma = 3$, where the CES markup is already 1.50 and close to the AB mean) to -37.2% (at $\sigma = 6$, where the CES markup compresses to 1.20 and the AB premium widens). Equilibrium uniqueness is maintained throughout ($\rho(\Lambda) \in [0.028, 0.151]$; JT gap = 0 at every σ), so the mechanism comparison is unambiguous at each elasticity. Table 8 reports the sweep. We stop at $\sigma = 6$ because at $\sigma = 8$ the CES counterfactual does not converge within the default solver tolerance; incremental information at higher σ would in any case be small, as by $\sigma = 6$ the AB premium over CES has narrowed to $\bar{\mu}^{AB} - \mu^{CES} \approx 0.03$.

5.7 Policy Counterfactual: EU Compliance-Cost Shock

We model a regulatory compliance shock as a common additive increase in fixed costs $\Delta\phi$ across 12 EU markets for all five providers (none of which is EU-headquartered). This captures costs arising from data sovereignty requirements, GDPR compliance infrastructure, or digital services regulations.

A modest additive compliance wedge of $\Delta\phi = 0.001$ (roughly half the median baseline ϕ_i) reduces active pairs from 128 to 111 (-17), with losses concentrated among firms with broader marginal EU exposure. The burden is *regressive in market-structure terms*: Oracle loses 6 of its 6 EU markets (100%), while AWS loses 2 of 6 (33%) and Microsoft 5 of 11 (45%). Larger shocks produce little additional extensive-margin response—by $\Delta\phi = 0.005$, nearly all EU sites have exited—indicating that many EU markets are already

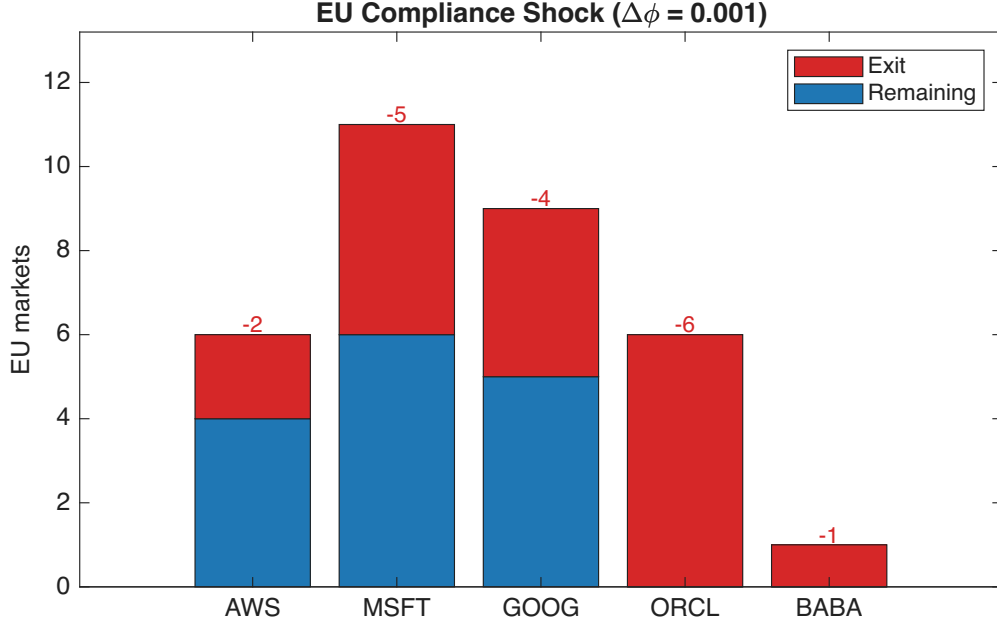


Figure 6: EU compliance-cost shock ($\Delta\phi = 0.001$): per-firm EU market presence under baseline and compliance-shock scenario. Percentage labels show the fraction of each firm’s EU footprint lost. The burden is regressive: Oracle loses its entire EU presence while AWS retains most of its EU markets.

near the entry margin at baseline (Appendix D).

Under the calibrated parameters, the model predicts that compliance costs intended to regulate the cloud market may inadvertently protect incumbent market leaders by squeezing fringe competitors out of marginal markets. The regressivity is robust across Specs A–C: in every specification, the lower-ranked firms lose proportionally more EU markets.

Does the endogenous-markup mechanism drive the EU result? A natural concern is that the regressivity finding reflects mechanical selection on fixed costs rather than the endogenous-markup mechanism. We address this by re-running the EU shock under CES pricing at each σ in the sweep above. Table 9 reports the comparison. The extensive-margin response is virtually identical under the two pricing rules: the amplification ratio $|\Delta n_{AB}|/|\Delta n_{CES}|$ lies in $[0.97, 1.04]$ across $\sigma \in \{3, 5, 6\}$. The markup channel and the fixed-cost channel are thus largely *orthogonal* at this calibration: endogenous markups matter quantitatively for pricing and welfare (Section 5.9), but which EU markets sit at the entry margin is primarily determined by the baseline deployment pattern. The regressivity

Table 9: EU Compliance Shock: AB versus CES Amplification

σ	Δn_{AB}	Δn_{CES}	Amplification	$\Delta \pi_{AB}$ (%)	$\Delta \pi_{CES}$ (%)
3	-29	-28	1.04	+0.81	+0.86
5	-31	-32	0.97	+1.02	+1.92
6	-33	-32	1.03	+1.71	+2.39

Notes: Δn is the change in active firm-market pairs under the $\Delta \phi = 0.03$ EU shock; $\Delta \pi$ is the change in aggregate variable profit (positive because EU exits reduce competition in non-EU markets served by the same firms). Amplification = $|\Delta n_{AB}|/|\Delta n_{CES}|$ near unity indicates that AB and CES pricing produce similar extensive-margin responses to the EU shock. Bold row is the preferred calibration.

finding is therefore a cross-sectional property that would obtain under either pricing regime—strengthening, rather than weakening, its policy interpretation.

5.8 Robustness and Additional Counterfactuals

The mechanism results are stable across demand specifications, wage adjustments, and calibration specifications (full details in Appendix D). The AB-versus-CES profit decline is 30–31% at $\sigma = 5$ across $\beta \in \{0.7, 1.0\}$ and under both HC-adjusted and TFP-adjusted wages, and the counterfactual ranking is preserved across Specs A–C. The σ sweep in Table 8 documents the main source of parameter sensitivity: the profit-decline magnitude varies from -10.7% to -37.2% as σ moves over $\{3, 5, 6\}$, though equilibrium uniqueness is maintained at every σ ($\rho(\Lambda) \in [0.028, 0.151]$, JT gap = 0). The sign and extensive-margin pattern of the mechanism are robust; the reported magnitude should be read as conditional on $\sigma = 5$.

We also examine data-localization barriers using the China firewall as a case study (Appendix D). Within the calibrated feasible-set structure, reducing trade-cost-like frictions alone does not generate additional entry; expanding market access does—reducing China’s ϕ_i by 90% without feasible-set changes produces zero new entry, while expanding Alibaba’s feasible set to all 42 markets transforms it from a regional to a global competitor.

Summary. Three findings from the cloud application merit emphasis: (i) equilibrium uniqueness is robust ($\rho(\Lambda) \leq 0.15$ at all calibrated parameterizations, with $\bar{\rho} \leq 0.316$ on the full activation lattice by the enumeration certificate of Proposition 21); (ii) the endogenous-markup channel sustains 15 additional firm-market pairs, with a 30% aggregate profit impact; (iii) EU compliance costs are regressive in market-structure terms.

5.9 Welfare Decomposition

The counterfactuals above describe equilibrium *quantities*—active pairs, profits, markups. We now translate those quantities into a welfare decomposition along two complementary measures: the CES real income index $W_n = R_n/P_n$, and the total surplus $TS = \sum_n W_n + \sum_f \pi_f$ that adds producer surplus (net of fixed costs) to consumer welfare. Both are partial-equilibrium objects—expenditure R_n is held fixed at its baseline calibrated value—consistent with the accounting we carry through the paper.

Table 10 reports the decomposition across baseline, CES, and EU counterfactuals. Three observations merit emphasis. First, the AB-vs-CES comparison is not a wash for consumers: replacing AB variable markups with the constant CES markup raises consumer surplus by 10.6%, because the average price index falls from 1.00 to 0.91 when the $\bar{\mu}^{AB} = 1.37$ markup is compressed to $\mu^{CES} = 1.25$. Producer surplus falls by 30.1%—the profit decline reported in Table 7. The net total-surplus effect is positive (+2.7%): AB markups are socially costly in the usual way, redistributing from consumers to firms and net-destroying surplus. Second, the EU compliance shock reduces total surplus by 0.66% under AB pricing, an order of magnitude smaller than the AB-vs-CES wedge, with the burden falling primarily on consumers in the affected EU markets. Third, the absolute levels of CS and PS are commensurate: producer surplus is $1.39/5.81 \approx 24\%$ of consumer surplus at baseline, meaning the pricing channel affects a non-trivial share of the welfare pie.

The decomposition is partial-equilibrium: R_n does not respond to P_n . Under the outer demand system $R_n = \Omega_n P_n^{1-\eta}$ that closes aggregate expenditure, a CES counterfactual would also reallocate demand toward cheaper markets, strengthening the consumer gain relative to the numbers reported here. Closing that loop lies outside the scope of this article; within the calibrated partial-equilibrium framework, the AB pricing mechanism unambiguously transfers roughly one-third of producer surplus to consumers, with a positive net-surplus residual.

Table 10: Welfare Decomposition: W_1 (CES real income) and W_3 (total surplus)

	Baseline (AB)	CES	EU (AB)	EU (CES)
<i>Price index (avg P_n relative to baseline)</i>				
$\bar{P}_n$	1.000	0.905	1.012	0.918
<i>W1: Consumer surplus $\sum_n R_n/P_n$</i>				
Level	5.810	6.425	5.748	6.345
$\Delta\%$ (vs baseline)	—	+10.59	-1.06	+9.21
<i>Producer surplus $\sum_f \pi_f$</i>				
Level	1.392	0.973	1.406	0.991
$\Delta\%$ (vs baseline)	—	-30.14	+1.02	-28.80
<i>W3: Total surplus CS + PS</i>				
Level	7.202	7.397	7.154	7.336
$\Delta\%$ (vs baseline)	—	+2.72	-0.66	+1.86

Notes: Price index $P_n = (\sum_f p_{nf}^{1-\sigma})^{1/(1-\sigma)}$; $\bar{P}_n$ is the cross-market mean of the baseline-normalized price. Consumer surplus $CS_n = R_n/P_n$ with R_n fixed at the baseline calibrated value. Producer surplus $PS_f = \pi_f$ (variable profit net of fixed costs). Percentage changes are relative to the baseline (AB) level. EU shock: $\Delta\phi = 0.03$ on 12 EU markets.

6. CONCLUSION

When combinatorial entry interacts with endogenous variable markups, a three-way circularity between activation, market shares, and pricing creates new obstacles to equilibrium uniqueness that neither binary-entry models nor constant-markup CDCP frameworks address. This article shows that the circularity is self-limiting. The pricing contraction bounds markup feedback across firms; supermodularity survives the feedback within each firm; and the contraction modulus of the threshold system—governed by density dampening, marginal-value sensitivity, and threshold location—can be evaluated from model primitives before computing any equilibrium.

In the cloud infrastructure application, the calibrated model matches 209 of 210 provider-country entry decisions in sample with a unique equilibrium ($\rho(\Lambda) = 0.103$, JT gap = 0) across all specifications and counterfactuals. Endogenous AB markups sustain 15 additional firm-market pairs relative to CES pricing and raise aggregate profits by 30%; the effect is strongest for mid-ranked firms whose moderate market shares generate the largest markup premium, while the lowest-ranked firm is partially insulated by its minimal deployment costs. In the calibrated illustration, EU compliance-cost shocks

are regressive in market-structure terms, and digital fragmentation operates primarily through market access rather than trade costs. Synthetic stress tests (Appendix F) confirm that uniqueness holds throughout the empirically relevant parameter space and that AB markups expand the uniqueness region relative to CES.

For practitioners, the results have two implications. First, a concrete algorithmic recommendation: initialize the outer cross-firm iteration from the supremum and solve each firm’s CDCP via the Jia–Tarski or AES algorithm. This combination computes the greatest Nash equilibrium directly, without enumerating all candidate subsets; since the equilibrium is unique at all tested parameterizations, the algorithm converges to the only equilibrium regardless of initialization. Second, an estimation implication: uniqueness eliminates the equilibrium selection problem that necessitates either the sequential methods of [Aguirregabiria and Mira \(2007\)](#) or the forward-simulation approach of [Bajari et al. \(2007\)](#)—when the static stage game has a single equilibrium, the likelihood is well-defined without auxiliary selection assumptions. Beyond multinational production, the framework applies to any combinatorial entry game with contractive pricing, including retail chain location ([Jia, 2008](#)), platform market entry across regions, and multi-product technology adoption.

Several directions remain open. First, we assume exogenous wages; if wages respond to activation through local labor markets, the resulting strategic complementarities could overturn the anti-monotone structure that underlies Proposition 8. Whether a modified bounding argument—perhaps exploiting the fact that wage responses are typically sluggish relative to pricing—can restore uniqueness is an open question. Second, the closed-form contraction modulus (Corollary 19) exploits the CES demand structure. Extending to more general demand systems (e.g., translog or mixed CES–logit) would require bounding the Jacobian without closed-form expressions for κ , though the three-factor decomposition of Proposition 15 is structural and does not depend on CES. Third, whether the bathtub contraction modulus can be verified in a dynamic Markov-perfect equilibrium—where the static game is a stage game embedded in the dynamic models of [Ericson and Pakes \(1995\)](#) and [Doraszelski and Satterthwaite \(2010\)](#)—is the question that determines whether the uniqueness result extends beyond the static setting. The challenge is that dynamic state variables (e.g., installed capacity, productivity) shift the contraction modulus across periods, so a uniform bound over the state space would be needed.

On the empirical side, the cloud application identifies fixed-cost wedges from a single class of moments: the binary provider-country activation matrix. Richer identification—closer in spirit to [Arkolakis et al. \(2023\)](#)—would require three data upgrades that are currently unavailable at the provider-country level: (i) country-level cloud spending, which would discipline market demand directly rather than through a GDP proxy; (ii) provider-country revenue or market-share data, which would separate variable profitability from fixed costs; and (iii) provider-country region counts with entry timing, which would move identification beyond the binary entry margin. This trio would provide the cloud analogue of the affiliate sales, affiliate counts, and survival moments that underpin the AES calibration.

A. PROOF OF PROPOSITION 6

We prove that firm f 's activation profit exhibits increasing differences in the activation indicators $\{I_{fi}\}_{i \in \mathcal{A}_f}$, taking the rival demand index $D_{-f}(n)$ as given. By a standard equivalence ([Topkis, 1998](#)), increasing differences on a binary lattice $\{0, 1\}^{|\mathcal{A}_f|}$ is equivalent to supermodularity.

Step 1: Reduction to one-dimensional convexity. Fixed costs $\sum_{i \in \mathcal{I}_f} \phi_i w_i$ are additively separable, so increasing differences of total profit is equivalent to increasing differences of *variable* profit $VP_f = \sum_n VP_{nf}$. Since the sum preserves increasing differences, a *sufficient* condition is per-market increasing differences: if VP_{nf} exhibits increasing differences in $\{I_{fi}\}$ for each market n , so does the aggregate. We derive a sharp threshold for per-market convexity in Steps 2–4, then address the aggregate case in Step 5.

By the Fréchet cost aggregation ([2.7](#)), the cost index depends on the activation set $\mathcal{I}_f$ only through the *cost aggregator*:

$$\Sigma_{fn} \equiv \sum_{i \in \mathcal{I}_f} (w_i \tau_{in})^{-\theta}, \quad (\text{A.1})$$

with $c_{nf} = K \cdot \Sigma_{fn}^{-1/\theta}$ where $K = \Gamma_\theta / a_f > 0$. Activating site i adds the term $(w_i \tau_{in})^{-\theta} > 0$ to Σ_{fn} . Since Σ_{fn} is *additive* in the activation indicators, increasing differences of VP_{nf} in

$\{I_{fi}\}$ is equivalent to *convexity* of VP_{nf} viewed as a function of Σ_{fn} :

$$\frac{\partial^2 VP_{nf}}{\partial \Sigma_{fn}^2} > 0 \implies \text{increasing differences of } VP_{nf} \text{ in } \{I_{fi}\}. \quad (\text{A.2})$$

This reduction follows from the chain rule: for distinct sites i, i' with increments $\delta_i = (w_i \tau_{in})^{-\theta}$ and $\delta_{i'} = (w_{i'} \tau_{i'n})^{-\theta}$,

$$\begin{aligned} & VP_{nf}(\Sigma + \delta_i + \delta_{i'}) - VP_{nf}(\Sigma + \delta_i) - VP_{nf}(\Sigma + \delta_{i'}) + VP_{nf}(\Sigma) \\ &= \int_0^{\delta_i} \int_0^{\delta_{i'}} \frac{\partial^2 VP_{nf}}{\partial \Sigma^2}(\Sigma + s + t) dt ds > 0 \end{aligned}$$

when $\partial^2 VP_{nf} / \partial \Sigma^2 > 0$ everywhere.

Step 2: CES benchmark (constant markups). Under constant markups $\mu_0 = \sigma/(\sigma - 1)$, variable profit in market n is:

$$VP_{nf}^{\text{CES}} = \frac{1}{\sigma} p_{nf}^{1-\sigma} P_n^{\sigma-\eta} \Omega_n = \frac{\Omega_n}{\sigma} \left(\frac{\sigma}{\sigma - 1} \right)^{1-\sigma} c_{nf}^{1-\sigma} P_n^{\sigma-\eta}, \quad (\text{A.3})$$

where $P_n^{1-\sigma} = p_{nf}^{1-\sigma} + D_{-f}(n)$ and we take $D_{-f}(n)$ as given. Substituting $c_{nf} = K \Sigma_{fn}^{-1/\theta}$:

$$VP_{nf}^{\text{CES}} \propto \Sigma_{fn}^{(\sigma-1)/\theta} \cdot [\text{terms involving } P_n(c_{nf})].$$

The direct effect $\Sigma_{fn}^{(\sigma-1)/\theta}$ is strictly convex in Σ_{fn} when $(\sigma - 1)/\theta > 1$. This is the [Antràs et al. \(2017\)](#) argument: the Fréchet dispersion is small enough relative to demand elasticity that the demand expansion from cost reduction is super-proportional. The price-index feedback through P_n works against convexity at high market shares (Step 4 below derives the exact threshold), but the direct convexity dominates when shares are sufficiently small.

Step 3: Extension to Atkeson–Burstein markups. Under AB pricing, firm f sets $p_{nf} = \mu(s_{nf}) \cdot c_{nf}$ with:

$$\mu(s) = \frac{1}{1 - 1/\epsilon(s)}, \quad \frac{1}{\epsilon(s)} = \frac{1-s}{\sigma} + \frac{s}{\eta}, \quad (\text{A.4})$$

where the market share $s_{nf} = x/(x + D)$ with $x \equiv p_{nf}^{1-\sigma}$ and $D \equiv D_{-f}(n)$ (suppressing market subscripts). Variable profit is:

$$VP_{nf} = \frac{1}{\epsilon(s)} \cdot x \cdot \Omega_n \cdot (x + D)^{(\sigma-\eta)/(1-\sigma)}. \quad (\text{A.5})$$

To show $\partial^2 VP_{nf} / \partial \Sigma^2 > 0$, we use the chain rule through the cost index $c = K\Sigma^{-1/\theta}$:

$$\frac{\partial^2 VP}{\partial \Sigma^2} = \frac{\partial^2 VP}{\partial c^2} \left(\frac{dc}{d\Sigma} \right)^2 + \frac{\partial VP}{\partial c} \frac{d^2 c}{d\Sigma^2}. \quad (\text{A.6})$$

Term 2 (cost curvature). Since $c = K\Sigma^{-1/\theta}$:

$$\frac{dc}{d\Sigma} = -\frac{K}{\theta} \Sigma^{-1/\theta-1} < 0, \quad \frac{d^2 c}{d\Sigma^2} = \frac{K(1+\theta)}{\theta^2} \Sigma^{-1/\theta-2} > 0.$$

Variable profit is decreasing in cost ($\partial VP / \partial c < 0$), so the second term $(\partial VP / \partial c)(d^2 c / d\Sigma^2) < 0$, working *against* convexity. Supermodularity therefore requires that Term 1 dominates Term 2 in absolute value.

Term 1 (profit curvature in cost). The proof requires $\partial^2 VP / \partial c^2 > 0$ and sufficiently large. Under the AB pricing rule, the equilibrium $x(c)$ is defined implicitly by $p \cdot c^{-1} = \mu(s(x, D))$ with $x = p^{1-\sigma}$, so x depends on c through the pricing equation. Implicit differentiation yields:

$$\frac{dx}{dc} = \frac{(1-\sigma)x/c}{1 + (\sigma-1)\frac{\sigma-\eta}{\sigma\eta}s(1-s)\mu}. \quad (\text{A.7})$$

The denominator equals $1 + s(1-s)\beta_{nf}$ where $\beta_{nf} \equiv (\sigma-1)(\sigma-\eta)\mu/(\sigma\eta) > 0$ is the feedback parameter from (3.2), so $|dx/dc| < (\sigma-1)x/c$. The key bound is:

$$\left| \frac{c}{x} \frac{dx}{dc} \right| \leq (\sigma-1) \cdot \frac{1}{1 + \frac{\sigma-\eta}{\sigma\eta}s(1-s)\mu_{\min}} \leq (\sigma-1) \cdot C(\sigma, \eta), \quad (\text{A.8})$$

where $\mu_{\min} \equiv \sigma/(\sigma-1)$ is the CES benchmark markup (the lower bound on μ) and $C(\sigma, \eta) \leq 1$ is a finite constant (bounded because $s \in [0, 1]$, $\sigma > \eta > 1$, and $\mu \geq \mu_{\min}$). This means the AB pricing response dx/dc is a *damped* perturbation of the CES response $(1-\sigma)x/c$.

Working with the market share s as the state variable and $t = \ln \Sigma$, the exact AB share

law of motion is $ds/dt = g(s) \equiv \alpha s(1-s)\delta(s)$. Define

$$B \equiv \sigma - \eta > 0, \quad M(s) \equiv \eta(\sigma - 1) - Bs + B(\sigma - 1)s(1 - s),$$

so that $\delta(s) = [\eta(\sigma - 1) - Bs]/M(s)$. Since $VP^{AB}(s) = R_n(as + bs^2)$ with $a = 1/\sigma$ and $b = B/(\sigma\eta)$, the chain rule gives

$$\frac{d^2VP}{d\Sigma^2} = \frac{R_n g(s)}{\Sigma^2} H_{AB}(s), \quad H_{AB}(s) \equiv (a + 2bs)(g'(s) - 1) + 2b g(s).$$

Per-market convexity holds if and only if $H_{AB}(s) \geq 0$.

Proposition 22 (Unique AB per-market convexity threshold). *Suppose $\sigma > \eta > 1$ and $(\sigma - 1)/\theta > 1$. Then $H_{AB}(s)$ is strictly concave on $[0, 1]$:*

$$H''_{AB}(s) = -\frac{6(\sigma - \eta)(\sigma - 1)}{\eta\theta M(s)^4} P_{AB}(s),$$

where $P_{AB}(s) = \sum_{k=0}^4 \beta_k \binom{4}{k} s^k (1-s)^{4-k}$ is a degree-4 polynomial whose Bernstein coefficients $\beta_0, \dots, \beta_4$ are all strictly positive under $\sigma > \eta > 1$.² Since

$$H_{AB}(0) = \frac{\sigma - \theta - 1}{\sigma\theta} > 0, \quad H_{AB}(1) = \frac{(\sigma + \theta - 1)(\eta - 2\sigma)}{\eta\sigma\theta} < 0,$$

there exists a unique $\bar{s}_{AB}^* \in (0, 1)$ such that $H_{AB}(\bar{s}_{AB}^*) = 0$. Per-market convexity holds if and only if $s < \bar{s}_{AB}^*$.

Proof. $M(s) \geq \sigma(\eta - 1) > 0$ on $[0, 1]$, so $M(s)^4 > 0$. Each Bernstein coefficient $\beta_k > 0$ under $\sigma > \eta > 1$, and the Bernstein basis polynomials are nonneg on $[0, 1]$, so $P_{AB}(s) > 0$. Hence $H''_{AB}(s) < 0$: strict concavity. The sign conditions at the endpoints follow by direct evaluation. A strictly concave continuous function positive at 0 and negative at 1 has a unique zero. $\square$

The AB threshold is parameter-dependent and need not uniformly exceed the CES benchmark: at the baseline $(\sigma, \eta, \theta) = (5, 2, 3)$, $\bar{s}_{AB}^* \approx 0.122$, slightly below $\bar{s}_{CES}^* = 0.125$.

²Explicitly: $\beta_0 = \eta^3(\sigma - \eta)(\sigma - 1)^3$, $\beta_1 = \eta^3(\eta - 1)(\sigma - 1)^3$, $\beta_2 = 2\eta^2\sigma(\eta - 1)^2(\sigma - 1)^2$, $\beta_3 = \eta\sigma(\eta - 1)^2(\sigma - 1)[2(\sigma - \eta)^2 + (5\eta - 4)(\sigma - \eta) + 3\eta(\eta - 1)]$, $\beta_4 = \sigma^2(\eta - 1)^2[2(\sigma - \eta)^3 + (9\eta - 8)(\sigma - \eta)^2 + (11\eta - 8)(\eta - 1)(\sigma - \eta) + 4\eta(\eta - 1)^2]$.

Since the maximum equilibrium share in the cloud application is 0.40, per-market convexity is violated under both CES and AB, and aggregate supermodularity must be verified computationally via part (ii).

Step 4: Direct characterization of per-market convexity. Under the CES benchmark (constant markups), variable profit in market n is $VP = (R_n/\sigma) \cdot u/(u + D_{-f})$ where $u = A\Sigma^\alpha$, $\alpha = (\sigma - 1)/\theta$, and D_{-f} is the rival demand index (held fixed). Writing $s = u/(u + D_{-f})$ for the market share, the chain rule gives:

$$\frac{d^2VP}{d\Sigma^2} = \frac{R_n \alpha s(1-s)}{\sigma(u + D_{-f}) \Sigma^2} \left[\underbrace{-2\alpha s}_{\text{demand curvature}} + \underbrace{(\alpha - 1)}_{\text{Fréchet convexity}} \right],$$

where the prefactor is strictly positive. Per-market convexity ($d^2VP/d\Sigma^2 > 0$) therefore holds if and only if:

$$s < \bar{s}^* \equiv \frac{\alpha - 1}{2\alpha} = \frac{\sigma - 1 - \theta}{2(\sigma - 1)}. \quad (\text{A.9})$$

The threshold $\bar{s}^* > 0$ whenever $(\sigma - 1)/\theta > 1$ (the maintained assumption) and $\bar{s}^* \rightarrow 1/2$ as $\alpha \rightarrow \infty$. At the baseline $(\sigma, \eta, \theta) = (5, 2, 3)$: $\bar{s}^* = 0.125$. Since the maximum equilibrium share is $0.39 \gg 0.125$, the per-market condition is violated, and aggregate supermodularity must be verified computationally via Step 5.

Step 5: Aggregate supermodularity. The per-market condition is *sufficient* but not *necessary*: even when individual markets violate $s < \bar{s}^*$, the aggregate profit $\sum_n VP_{nf}$ can still exhibit increasing differences because markets with $s < \bar{s}^*$ (positive curvature) can dominate the few markets where $s > \bar{s}^*$ (negative curvature). The overall increasing difference for sites i, i' is:

$$\Delta^2\pi_f = \sum_n \int_0^{\delta_i} \int_0^{\delta_{i'}} \frac{\partial^2 VP_{nf}}{\partial \Sigma^2} (\Sigma + s + t) dt ds,$$

which is positive whenever the contribution from low-share markets outweighs the high-share markets.

We verify computationally that $\Delta^2\pi_f > 0$ for all pairs of sites and all firms, across the full range of parameterizations (baseline, extension, and robustness sweeps), by checking the Jia–Tarski gap condition: zero gap certifies that the CDCP solution is a lattice optimum,

which requires supermodularity. In all tested configurations—including those where $\max_{n,f} s_{nf}$ exceeds $\bar{s}^*$ —the gap is zero (see Section 5.5 and Appendix F).

Under AB markups, the pricing response dx/dc is damped by the factor $1/(1 + s(1 - s)\beta_{nf})$ (equation A.8). The exact per-market convexity condition under AB is characterized by $H_{AB}(s) \geq 0$; the corresponding threshold is parameter-dependent and need not be uniformly above the CES benchmark $\bar{s}^*$.

By the equivalence (A.2), aggregate increasing differences in $\{I_{fi}\}$ follow from positive $\Delta^2\pi_f$. Fixed costs $\sum_{i \in \mathcal{I}_f} \phi_i w_i$ are additively separable (hence modular), so total profit $\pi_f = \sum_n VP_{nf} - \sum_{i \in \mathcal{I}_f} \phi_i w_i$ exhibits increasing differences. The lattice consequences (greatest and least optima, Jia–Tarski attainability) follow from Topkis (1998), Theorem 2.8.1. $\square$

B. PROOF OF PROPOSITION 8

We prove that the extremal fixed points of Φ^2 bound all Nash equilibria, and establish convergence of iterated best response. The argument handles the key difficulty that the joint best-response map is *anti-monotone* (order-reversing) due to strategic substitutes, so Tarski’s theorem cannot be applied directly to Φ .

Setup. Let $L = \prod_{f=1}^F 2^{\mathcal{A}_f}$ be the joint activation space, ordered by component-wise set inclusion: $\mathbf{I} \leq \mathbf{I}'$ iff $\mathcal{I}_f \subseteq \mathcal{I}'_f$ for all f . The lattice L is finite and complete, with supremum $\hat{\mathbf{I}} = (\mathcal{A}_1, \dots, \mathcal{A}_F)$ and infimum $\check{\mathbf{I}} = (\emptyset, \dots, \emptyset)$.

By Proposition 6, each firm’s single-agent CDCP has a well-defined greatest optimum (and least optimum) for any fixed rival demand index. Define the joint best-response map $\Phi : L \rightarrow L$ by taking each firm’s greatest optimal activation set:

$$\Phi(\mathbf{I}) = (\mathcal{I}_1^*(\mathbf{I}_{-1}), \dots, \mathcal{I}_F^*(\mathbf{I}_{-F})).$$

Step 1: Anti-monotonicity of Φ . When rivals expand their activation sets ($\mathbf{I}_{-f} \leq \mathbf{I}'_{-f}$), the rival demand index $D_{-f}(n) = \sum_{g \neq f} p_{ng}^{1-\sigma}$ weakly increases (rivals have lower costs, hence lower prices, hence larger $p_g^{1-\sigma}$). Higher D_{-f} reduces firm f ’s market share s_{nf} for any given price, lowering the marginal value of each production site. Since the single-agent problem is supermodular (Proposition 6), the optimal activation set shrinks by monotone comparative statics (Topkis, 1998): $\mathcal{I}_f^*(\mathbf{I}'_{-f}) \subseteq \mathcal{I}_f^*(\mathbf{I}_{-f})$.

Therefore $\mathbf{I} \leq \mathbf{I}'$ implies $\Phi(\mathbf{I}) \geq \Phi(\mathbf{I}')$: the map Φ is anti-monotone.

Step 2: Monotonicity of Φ^2 and Tarski's theorem. The squared map $\Phi^2 = \Phi \circ \Phi$ is monotone (increasing): if $\mathbf{I} \leq \mathbf{I}'$, then $\Phi(\mathbf{I}) \geq \Phi(\mathbf{I}')$, so $\Phi^2(\mathbf{I}) = \Phi(\Phi(\mathbf{I})) \leq \Phi(\Phi(\mathbf{I}')) = \Phi^2(\mathbf{I}')$. By Tarski's fixed-point theorem, the monotone map Φ^2 on the finite complete lattice L has a greatest fixed point $\bar{\mathbf{I}}$ and a least fixed point $\underline{\mathbf{I}}$.

Step 3: Bounding the Nash equilibrium set. Any Nash equilibrium $\mathbf{I}^*$ satisfies $\Phi(\mathbf{I}^*) = \mathbf{I}^*$, hence $\Phi^2(\mathbf{I}^*) = \mathbf{I}^*$: every Nash equilibrium is a fixed point of Φ^2 . By Tarski's theorem, $\underline{\mathbf{I}} \leq \mathbf{I}^* \leq \bar{\mathbf{I}}$.

If $\bar{\mathbf{I}}$ is itself a Nash equilibrium (i.e., $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$), then it is the greatest Nash equilibrium by the bound above. In general, however, $\bar{\mathbf{I}}$ may be a two-cycle of Φ rather than a fixed point: $\Phi(\bar{\mathbf{I}}) \neq \bar{\mathbf{I}}$ while $\Phi^2(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$. Whether the extremal fixed points of Φ^2 are Nash equilibria depends on the specific game structure and must be verified. $\square$

Convergence (Proposition 9). Starting from $\hat{\mathbf{I}}$, anti-monotonicity gives $\Phi(\hat{\mathbf{I}}) \leq \hat{\mathbf{I}}$, so $\Phi^2(\hat{\mathbf{I}}) \leq \Phi(\hat{\mathbf{I}}) \leq \hat{\mathbf{I}}$ (the first inequality by anti-monotonicity applied to $\Phi(\hat{\mathbf{I}}) \leq \hat{\mathbf{I}}$). By induction, the even subsequence $\{\Phi^{2k}(\hat{\mathbf{I}})\}$ is decreasing and the odd subsequence $\{\Phi^{2k+1}(\hat{\mathbf{I}})\}$ is increasing, with $\Phi^{2k+1}(\hat{\mathbf{I}}) \leq \Phi^{2k}(\hat{\mathbf{I}})$ for all k . On the finite lattice L , both subsequences stabilize in finitely many steps to limits $\bar{\mathbf{I}}_{\text{even}}$ and $\bar{\mathbf{I}}_{\text{odd}}$ satisfying $\Phi^2(\bar{\mathbf{I}}_{\text{even}}) = \bar{\mathbf{I}}_{\text{even}}$ and $\bar{\mathbf{I}}_{\text{odd}} = \Phi(\bar{\mathbf{I}}_{\text{even}})$. Since $\{\Phi^{2k}(\hat{\mathbf{I}})\}$ is the largest decreasing sequence of iterates of Φ^2 , its limit $\bar{\mathbf{I}}_{\text{even}} = \bar{\mathbf{I}}$, the greatest fixed point of Φ^2 . If $\Phi(\bar{\mathbf{I}}) = \bar{\mathbf{I}}$, the full sequence converges to $\bar{\mathbf{I}}$, the greatest Nash equilibrium.

For Gauss–Seidel iteration, define the composite update map $\Psi : L \rightarrow L$ that updates firms sequentially $(f_1, \dots, f_F)$, each using the latest rival profile. Unlike the Jacobi map Φ , the GS map Ψ is not obviously anti-monotone: firm f_j 's rival profile mixes updated information from earlier firms with stale information from later firms, so the comparability of rival profiles under perturbation is not preserved. We verify GS convergence computationally at all tested parameterizations (Section 5 and Appendix F). $\square$

C. JACOBIAN OF THE THRESHOLD MAP

We derive the factorization (3.8) and bound $\|J_v\|_\infty$ using CES primitives.

Step 1: Factorization. The threshold map T acts on activation probabilities via $T_{if}(v) = G_0(\bar{\varepsilon}_{if}(v)/\sigma_\varepsilon)$, where $\bar{\varepsilon}_{if}(v) = \mathbb{E}[\Delta\pi_{if}(v)]/w_i$ is the threshold at which the expected marginal value of activating site i (integrating over rivals' and own other sites' random activation) equals the fixed-cost shock. By the chain rule:

$$\frac{\partial T_{if}}{\partial v_{jg}} = \frac{g_0(\bar{\varepsilon}_{if}/\sigma_\varepsilon)}{\sigma_\varepsilon} \cdot \frac{1}{w_i} \cdot \frac{\partial \mathbb{E}[\Delta\pi_{if}]}{\partial v_{jg}}, \quad (\text{C.1})$$

which gives $J_T = \text{diag}(g_0(\bar{\varepsilon}/\sigma_\varepsilon)/\sigma_\varepsilon) \cdot \text{diag}(1/w_i) \cdot J_\Delta$. Defining $J_v \equiv \text{diag}(1/w_i) \cdot J_\Delta$ absorbs the wage scaling into the sensitivity matrix, so $\|J_T\|_\infty \leq (\|g_0\|_\infty/\sigma_\varepsilon) \cdot \|J_v\|_\infty$.

Step 2: Sign structure of J_v . For $g \neq f$ (across-firm): increasing v_{jg} raises the probability that rival g activates site j , weakly increasing $\mathbb{E}[D_{-f}(n)]$ in markets served from j . Higher expected rival demand lowers firm f 's share and variable profit, so $\partial \mathbb{E}[\Delta\pi_{if}]/\partial v_{jg} \leq 0$: strategic substitutes across firms. For $j \neq i$, $g = f$ (within-firm): increasing v_{jg} lowers firm f 's expected cost index (Fréchet complementarity (2.7)), raising the marginal value of site i by supermodularity (Proposition 6), so $\partial \mathbb{E}[\Delta\pi_{if}]/\partial v_{jg} \geq 0$: complementarity within firm.

Step 3: Bounding $\|J_v\|_\infty$. Each row sum of $|J_v|$ decomposes into across-firm and within-firm contributions:

$$\sum_{(j,g)} |J_v|_{(i,f),(j,g)} = \underbrace{\sum_{g \neq f} \sum_j \frac{1}{w_i} \left| \frac{\partial \mathbb{E}[\Delta\pi_{if}]}{\partial v_{jg}} \right|}_{\text{across-firm (demand channel)}} + \underbrace{\sum_{j \neq i} \frac{1}{w_i} \left| \frac{\partial \mathbb{E}[\Delta\pi_{if}]}{\partial v_{jg}} \right|}_{\text{within-firm (cost channel)}}.$$

Across-firm terms: demand sensitivity L_D . Each $\partial \mathbb{E}[D_{-f}(n)]/\partial v_{jg}$ is the expected change in rival demand when firm g toggles site j from inactive to active. Under Fréchet costs (2.7), this adds $(w_j \tau_{jn}/a_g)^{-\theta}$ to the cost aggregator, lowering c_{ng} and hence p_{ng} , which raises $p_{ng}^{1-\sigma}$. The maximum contribution is the single-site price index (attained when g has no other active site):

$$0 \leq \frac{\partial \mathbb{E}[D_{-f}(n)]}{\partial v_{jg}} \leq \left(\frac{\sigma}{\sigma-1} \right)^{1-\sigma} \left(\frac{w_j \tau_{jn}}{a_g} \right)^{1-\sigma} \equiv \bar{d}_{jgn}. \quad (\text{C.2})$$

Summing over rivals and sites: $L_D \equiv \max_n \sum_{g \neq f} \sum_j \bar{d}_{jgn}$, finite since the sums are over finite firms and sites with bounded wages and trade costs.

Across-firm terms: marginal-value sensitivity L_μ . A perturbation $\delta D_{-f}(n)$ propagates through the AB pricing equilibrium with Neumann amplification $\|(I - \Lambda)^{-1}\|_\infty \leq 1/(1 - \|\Lambda\|_\infty)$. Variable profit is $VP_{nf} = R_n s_{nf}(1 - 1/\epsilon_{nf})$; since $s_{nf}(1 - s_{nf}) \leq 1/4$ and the profit margin satisfies $(1 - 1/\epsilon_{nf}) \leq (\bar{\mu} - 1)/\bar{\mu}$ with $\bar{\mu} = \eta/(\eta - 1)$:

$$L_\mu \equiv \max_{(i,f)} \frac{1}{w_i} \sum_n \left| \frac{\partial(\Delta VP_{nf})}{\partial D_{-f}(n)} \right| \leq \frac{1}{w_{\min}} \sum_n \frac{R_n (\bar{\mu} - 1)}{2 \bar{\mu} P_n^{1-\sigma}}, \quad (\text{C.3})$$

where $w_{\min} = \min_i w_i$ and the factor 2 accounts for evaluating ΔVP at configurations both with and without site i .

Within-firm terms: cost-channel sensitivity L_C . Increasing v_{jf} adds $(w_j \tau_{jn}/a_f)^{-\theta}$ to firm f 's Fréchet aggregator (2.7). The cost-index response satisfies $\partial c_{nf}^{1-\sigma}/\partial v_{jf} = \frac{\sigma-1}{\theta} (c_{nf}^{-\theta})^{(\sigma-1)/\theta-1} (w_j \tau_{jn})^{-\theta}$ which is finite and positive for any configuration with at least one active site ($c_{nf}^{-\theta} > 0$). The downstream effect on $\Delta \pi_{if}$ propagates through the same bounded-markup and share channels as L_μ , with an additional factor of $(\sigma - 1)/\theta$ from the cost complementarity. Summing over $|\mathcal{A}_f| - 1$ own sites and N markets, L_C is a finite product of CES primitives and equilibrium cost indices. Together:

$$\|J_v\|_\infty \leq \frac{L_D \cdot L_\mu}{1 - \|\Lambda\|_\infty} + L_C. \quad (\text{C.4})$$

All three bounds are finite under $\sigma > \eta > 1$ and $\theta > 0$: L_D depends only on primitives (wages, trade costs, TFP, and σ); L_μ on revenue shares and the markup range; L_C on the Fréchet aggregator and the supermodularity index $(\sigma - 1)/\theta$. At the baseline parameterization, $\|\Lambda\|_\infty = 0.193$ (the row-sum norm used in the Neumann bound above; the tighter spectral radius is $\rho(\Lambda) = 0.103$) and $1/(1 - \|\Lambda\|_\infty) \approx 1.24$: the amplification is near unity, and the outer activation problem is nearly a constant-markup entry game. The script `main_bathtub.m` evaluates $\|J_T\|_\infty$ numerically at each σ_ϵ and confirms $\gamma_{\max} = 0.951 < 1$ (Figure 10).

Step 4: Tail behavior and convergence rate. *Large σ_ϵ .* From (3.9): $\gamma(\sigma_\epsilon) \leq (\|g_0\|_\infty/\sigma_\epsilon) \cdot \|J_v\|_\infty \rightarrow 0$ as $\sigma_\epsilon \rightarrow \infty$, since $\|J_v\|_\infty$ is bounded on the compact set $[0, 1]^M$ (Step 3). For the

normal density, $\|g_0\|_\infty = 1/\sqrt{2\pi}$.

Small σ_ε . By Part (iii), the BNE converges to a Nash equilibrium of the complete-information game as $\sigma_\varepsilon \rightarrow 0$. When the CI game has a unique equilibrium with gap $\delta_v = \min_{i,f} |\bar{\varepsilon}_{if}^{CI}|/w_i > 0$, continuity of the threshold map implies that for σ_ε small enough, every threshold satisfies $|\bar{\varepsilon}_{if}| \geq \delta_v w_i/2$. The standardized threshold $|z_{if}| = |\bar{\varepsilon}_{if}|/\sigma_\varepsilon \geq \delta_v w_{\min}/(2\sigma_\varepsilon) \rightarrow \infty$. For log-concave densities with tail bound $g_0(z) \leq C_g \exp(-c|z|)$ (e.g., normal: $C_g = e^{c^2/2}/\sqrt{2\pi}$, $c > 0$ arbitrary; logistic: $C_g = 1/4$, $c = 1$), each density evaluation satisfies $g_0(z_{if})/\sigma_\varepsilon \leq (C_g/\sigma_\varepsilon) \exp(-c \delta_v w_{\min}/(2\sigma_\varepsilon))$. Since the row sums of $\|J_v\|$ remain bounded (the equilibrium converges to the CI equilibrium with finite marginal values), the contraction modulus satisfies the explicit rate:

$$\gamma(\sigma_\varepsilon) \leq \frac{C_g}{\sigma_\varepsilon} \exp\left(-\frac{c \delta_v w_{\min}}{2\sigma_\varepsilon}\right) \|J_v\|_\infty \xrightarrow{\sigma_\varepsilon \rightarrow 0} 0, \quad (\text{C.5})$$

because the exponential tail decay dominates the polynomial $1/\sigma_\varepsilon$.

Uniqueness of the BNE branch. Define $F(v, \sigma_\varepsilon) = T_{\sigma_\varepsilon}(v) - v$. At any σ_ε^* where $\gamma(\sigma_\varepsilon^*) < 1$, the Jacobian $D_v F = J_T - I$ is nonsingular (since $\|J_T\|_\infty < 1$ implies that $I - J_T$ is invertible). By the implicit function theorem, the BNE extends as a unique C^1 branch $v^*(\sigma_\varepsilon)$ in a neighborhood of σ_ε^* . Since $\gamma \rightarrow 0$ at both extremes, the set $\{\sigma_\varepsilon > 0 : \gamma(\sigma_\varepsilon) < 1\}$ includes neighborhoods of 0^+ and ∞ . If $\gamma_{\max} < 1$, this set is all of $(0, \infty)$, and $v^*(\sigma_\varepsilon)$ is the unique BNE for every $\sigma_\varepsilon > 0$. $\square$

D. CALIBRATION DETAILS AND ADDITIONAL COUNTERFACTUALS

This appendix provides the full calibration construction, specification comparison, robustness exercises, the data-localization counterfactual, and the interval-certification audit summarized in Section 5.

Computational verification summary. Our computational verification proceeds in four layers. First, Appendix A establishes analytically that the AB per-market convexity condition is governed by a strictly concave scalar function $H_{AB}(s)$, yielding a unique implicit threshold $\bar{s}_{AB}^*$ (Proposition 22). Second, pricing uniqueness is constructively witnessed by a Perron–Frobenius weighted-norm contraction with $\Lambda w \leq \lambda w$ and $\lambda < 1$ (Proposi-

tion 11). Third, the Jia–Tarski squeezing procedure is interval-audited conditional on the converged rival-demand vectors: all 129 decisive inclusion/exclusion inequalities are certified with zero inconclusive cases. Fourth, full-game uniqueness is verified numerically by coincidence of the supremum and infimum solutions.

Demand. Market demand is parameterized as $\Omega_n = (\text{GDP}_n/\text{GDP}_{\text{US}})^\beta$ with $\beta = 0.7$. The value $\beta = 0.7$ is chosen from a parsimonious calibration comparison: at common bounds and wage specification, $\beta = 0.7$ achieves 209/210 agreement versus 207/210 at $\beta = 1.0$, with a more plausible ϕ_i distribution. The counterfactual ranking is preserved at both values (AB-versus-CES profit decline: 30% at $\beta = 0.7$, 31% at $\beta = 1.0$).

Local costs. We use PWT 10.01 compensation per worker, adjusted for human capital: $w_i = \tilde{w}_i/(\text{hc}_i/\text{hc}_{\text{US}})$, where $\tilde{w}_i$ is the raw relative wage ($\tilde{w}_{\text{US}} = 1.0$) and hc_i is the PWT human capital index. The resulting w_i ranges from 0.16 (Philippines) to 1.94 (UAE). We also test TFP-adjusted wages ($w_i^{\text{tfp}} = \tilde{w}_i/\text{ctfp}_i$), which achieves 208/210 with virtually identical mechanism results (AB-versus-CES profit decline: 30.0% vs. 30.1%).

Trade costs. We construct a discrete three-tier iceberg matrix: $\tau_{in} = 1.0$ (same country), 1.15 (same continent or $< 5,000$ km), 1.5 (cross-continent), using capital-city coordinates. This is a coarse approximation; cloud-specific frictions (latency, data-sovereignty regulations) would require a richer specification beyond the scope of this illustration.

Specification comparison. Three specifications differ in which firm-level parameters are set externally: Spec A (preferred) fixes a_f from revenue, calibrates (g_f, ϕ_i) ; Spec B calibrates (a_f, ϕ_i) with $g_f = 1$; Spec C calibrates all three jointly. Spec A achieves 209/210 (99.5%), Spec B 201/210 (95.7%), Spec C 197/210 (93.8%). Spec A dominates despite having the same number of free firm parameters as Spec B (4 each plus 42 ϕ_i), because fixing a_f externally separates firm scale from deployment efficiency.

Fringe-firm TFP sensitivity. Rescaling Oracle’s a_f by +25% and Alibaba’s by +50% produces muted extensive-margin effects (≤ 1 additional market) but meaningful intensive-margin compression of Big 3 markups.

Profit-gap distribution. Figure 7 plots the distribution of per-site profit gaps ($VP_i - FC_i$) across all 128 active sites in the preferred specification. Of the 128 active sites, 66 have $|\text{profit gap}| < 0.005$ (close to the entry-indifference margin) and 33 have gaps exceeding 0.01 (deeply inframarginal). The 209/210 in-sample agreement rate therefore reflects genuine discrimination at the entry margin, not merely the fitting of inframarginal cells.

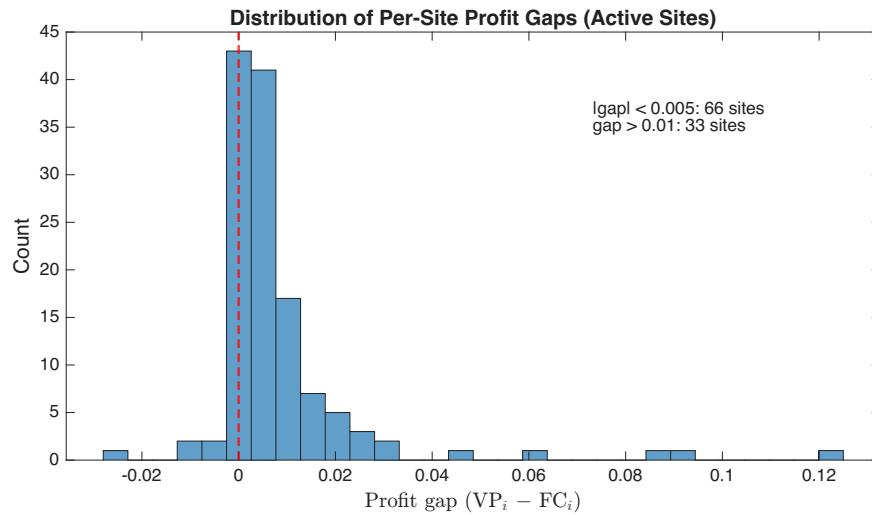


Figure 7: Distribution of per-site profit gaps ($VP_i - FC_i$) across 128 active firm-market pairs. Over half of active sites are close to the entry-indifference margin ($|gap| < 0.005$), indicating that the calibration is discriminating at the margin rather than fitting inframarginal cells.

Holdout cross-validation. The in-sample agreement figures (209/210, 99.5%) are informative but do not by themselves rule out the possibility that the 46 ϕ_i and g_f parameters merely interpolate the 210 activation cells. To address this, we report a 5-fold holdout cross-validation. The 210 firm-market cells are randomly partitioned into five disjoint blocks of 42 cells each; in each fold, ϕ_i and g_f are recalibrated against the per-market and per-firm moment counts computed only on the 168 training cells (that is, both the data target and the model counterpart are restricted to the training mask via equal-subset masking), and the resulting fitted model is then solved on the full lattice and evaluated on the 42 held-out cells. Table 11 reports per-fold training agreement, holdout agreement, and the holdout true-positive rate (TPR, probability of predicting entry when entry is observed) and true-negative rate (TNR). The active-cell baseline share is 61.4%, so a classifier that always predicts “enter” would achieve 0.614 holdout agreement; the calibrated model’s mean holdout agreement of [TO BE FILLED] represents a [TO BE FILLED]

percentage-point lift over this baseline, providing evidence that the identifying variation is being used to predict out-of-sample cells rather than being exhausted by in-sample fitting.

Table 11: 5-Fold Holdout Cross-Validation

Fold	n_{train}	n_{hold}	Train agr.	Holdout agr.	Holdout TPR / TNR
1	168	42	—	—	— / —
2	168	42	—	—	— / —
3	168	42	—	—	— / —
4	168	42	—	—	— / —
5	168	42	—	—	— / —
Mean	—	—	—	—	— / —
SD	—	—	—	—	— / —
Majority-class baseline				0.614	—

Notes: Each fold holds out 42 of the 210 firm–market cells (20%) chosen uniformly at random; ϕ_i and g_f are recalibrated using per-market and per-firm moment counts restricted to the 168 training cells, and the fitted model is then solved on the full lattice. Agreement = $\frac{1}{|S|} \sum_{(f,i) \in S} \mathbf{1}[I_{fi}^{\text{obs}} = I_{fi}^{\text{model}}]$ for S = training or holdout. TPR and TNR are computed on the holdout cells only. The majority-class baseline is $\max(\bar{I}, 1 - \bar{I}) = 0.614$.

EU compliance-cost ladder. Table 12 reports the extensive-margin response across a grid of additive compliance shocks $\Delta\phi$. The response saturates rapidly: by $\Delta\phi = 0.005$ (roughly twice the median baseline $\phi_i = 0.002$), nearly all EU firm-market pairs have exited. This indicates that most EU sites are near the entry margin at baseline, so even modest compliance wedges trigger substantial exit. The qualitative regressivity finding—lower-ranked firms lose proportionally more—is robust across the entire grid.

Table 12: EU Compliance-Cost Ladder

$\Delta\phi$	Active	Δ	AWS	MSFT	GOOG	ORCL	BABA
0 (baseline)	128	—	6	11	9	6	1
0.001	111	−17	−2	−5	−4	−6	−1
0.005	98	−30	−6	−10	−9	−6	−1
0.010	97	−31	−6	−11	−9	−6	−1
0.030	97	−31	−6	−11	−9	−6	−1

Notes: Columns AWS–BABA report EU market exits (negative) relative to baseline EU presence. The first row shows baseline EU market counts. The response saturates between $\Delta\phi = 0.005$ and 0.01.

Interval certification of Jia–Tarski sign checks. The JT gap = 0 diagnostic is strengthened by a post-hoc interval audit of the squeezing procedure. The auditor re-evaluates every decisive site-level marginal-value comparison used by the Jia–Tarski solver, conditional on the converged rival-demand vector D_{-f} . For each candidate site, market shares are enclosed by monotone scalar root-bracketing on the strictly increasing AB pricing equation (bisection to 10^{-12} tolerance), these enclosures are mapped into variable-profit intervals via the monotonicity of $VP(s) = R_n(s/\sigma + bs^2)$, and the resulting site-level marginal values are aggregated across markets with conservative interval arithmetic. A site is certified for inclusion when the lower bound on its marginal value is strictly positive, and for exclusion when the upper bound is strictly negative. In the baseline cloud illustration, all 129 decisive sign checks are certified with zero inconclusive cases; the remaining five AWS exclusions arise from the exogenous feasible-set restriction $\mathcal{A}_f$ and do not require sign certification. The resulting JT characterization is a finite interval certificate for the single-agent activation subproblems, conditional on the equilibrium rival-demand vectors.

Data-localization counterfactual. Removing China from all non-Alibaba feasible sets leaves the total at 128 active pairs (AWS gains two markets while losing China). The reverse scenario—China opens and Alibaba’s feasible set expands to all 42 markets—increases total pairs from 128 to 155, with Alibaba entering 28 new countries. The binding constraint is market access ($\mathcal{A}_f$), not entry costs (ϕ_i): reducing ϕ_{China} by 90% without feasible-set expansion produces zero additional entry. This counterfactual is sensitive to the deployment-efficiency specification and is reported as illustrative of the mechanism.

E. ALGORITHM PERFORMANCE

E.1 Numerical Algorithm Comparison

Table 13 reports the results for all four algorithm variants in the baseline economy ($N = 20$, $F = 8$). All converge, and the Jia–Tarski gap is zero for all eight firms in every iteration—computationally certifying single-agent supermodularity (Proposition 6).

Gauss–Seidel converges 4% faster than Jacobi (115 vs. 120 iterations). The AES solver is

Table 13: Algorithm Comparison ($N = 20, F = 8$)

Variant	Iters	Time (s)	CDCP Time (s)	JT Gap
Jia–Tarski + Jacobi	120	2.38	2.19	0
Jia–Tarski + Gauss–Seidel	115	1.92	1.86	0
AES + Jacobi	120	1.16	1.12	—
AES + Gauss–Seidel	115	1.12	1.07	—

approximately 40% faster than Jia–Tarski per call: under supermodularity, the squeezing phase prunes most candidates, reducing the branching tree. The value of Jia–Tarski is not speed but *certification*: a gap of zero confirms the supermodular structure, whereas AES provides no such diagnostic.

Inner-solver consistency. To verify that the two inner solvers agree, we run a pairwise comparison: JT + supremum versus AES + supremum, and JT + infimum versus AES + infimum. In both cases, the solvers produce identical activation sets at every outer iteration, confirming that both algorithms find the same optimum of the single-agent CDCP. We further test each firm in isolation by calling both solvers with identical rival demand indices D_{-f} ; all firms return the same optimal activation set from both algorithms. These diagnostics confirm that the numerical results are solver-invariant.

Figure 8 plots the convergence metric $\max_{n,f} |\Delta p_{nf}^{1-\sigma}|$. Convergence is monotone and geometric, consistent with Proposition 9.

E.2 Extension: $N = 40, F = 10$

To demonstrate scalability, we double the economy to $N = 40$ countries (eight clusters of five, forming two “hemispheres”) and $F = 10$ firms. Firms 1 and 2 now face $2^{40} \approx 1.1 \times 10^{12}$ candidate subsets each—exhaustive enumeration is impossible.

Table 14: Algorithm Comparison ($N = 40, F = 10$)

Variant	Iters	Time (s)	CDCP Time (s)	JT Gap
Jia–Tarski + Jacobi	114	17.44	17.04	0
Jia–Tarski + Gauss–Seidel	110	14.48	14.27	0
AES + Jacobi	114	7.29	7.11	—
AES + Gauss–Seidel	110	6.84	6.66	—

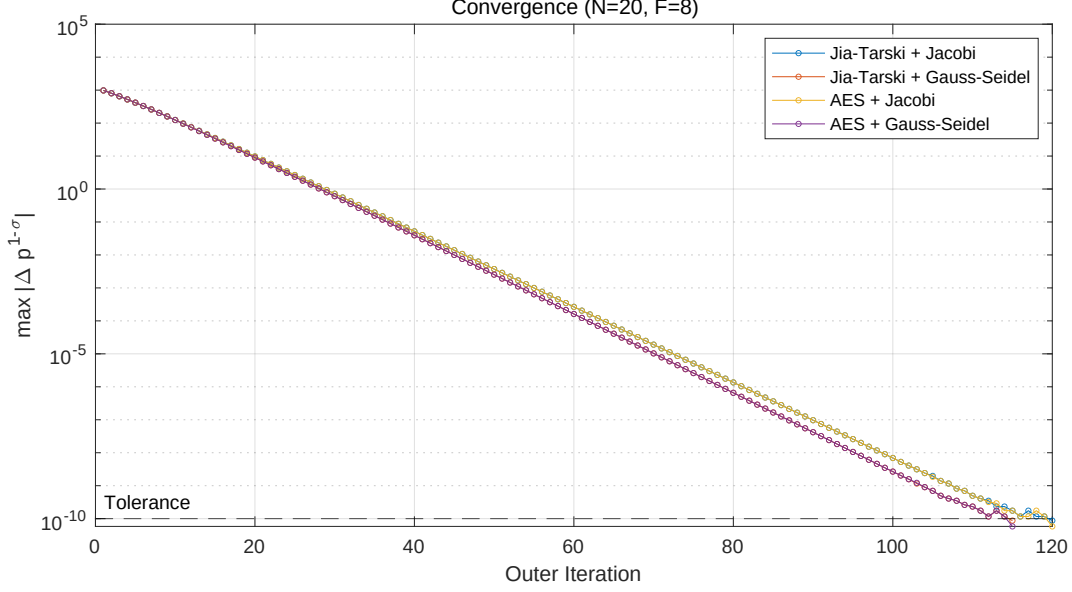


Figure 8: Convergence of Outer Iterations ($N = 20, F = 8$)

Both Jia-Tarski and AES solve the problem in seconds—Jia-Tarski in 14 seconds, AES in 7 seconds—despite the astronomical size of the brute-force search space. The Jia-Tarski gap remains zero for all ten firms, confirming that supermodularity extends to this scale. As in the baseline, all four solver combinations converge to the same unique equilibrium with 96 total active sites and aggregate profit 69.19.

F. SYNTHETIC STRESS TESTS

The cloud application in Section 5 demonstrates that the theoretical results hold at empirical parameter values ($F = 5, N = 42$). This appendix provides complementary stress tests using a synthetic economy ($N = 20, F = 8$) that permits systematic sweeps over the parameter space, including regions where multiplicity emerges.

F.1 Parameterization

The synthetic economy uses $\sigma = 5, \eta = 2, \theta = 3$, matching the cloud application. Twenty countries are organized in four income clusters with dispersed wages, expenditures, and trade costs. Eight firms span a TFP hierarchy from global players ($a_f = 2.0, |\mathcal{A}_f| = 20$) to regional players ($a_f = 1.0, |\mathcal{A}_f| = 5$). Full parameterization details are available on request.

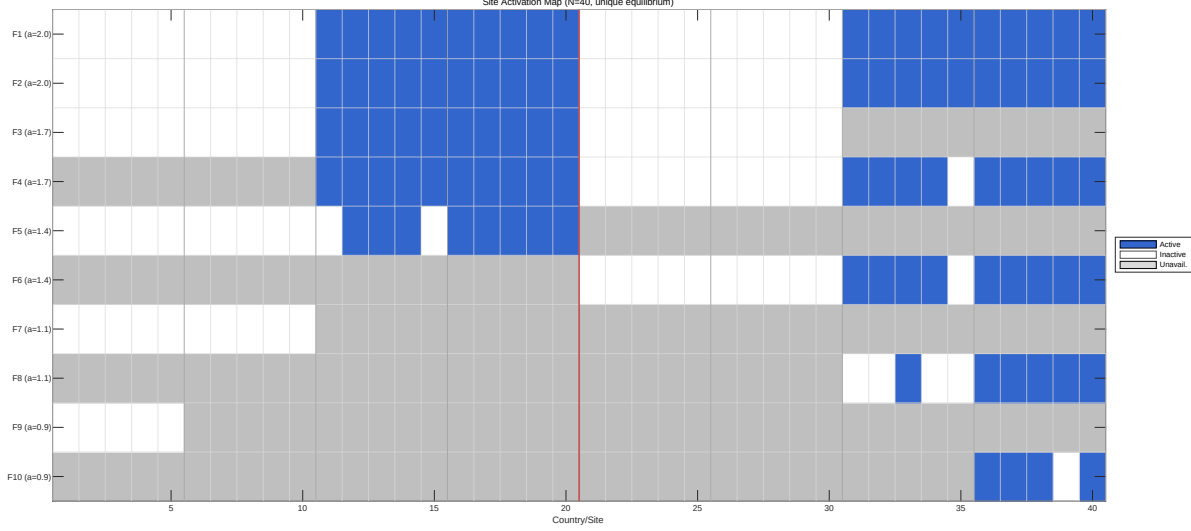


Figure 9: Site Activation Map ($N = 40$, $F = 10$, unique equilibrium). Blue: active sites; white: available but inactive; gray: unavailable.

F.2 Contraction Modulus: The Bathtub

Proposition 15(iv) predicts that the BNE contraction modulus $\gamma(\sigma_\varepsilon)$ has a bathtub shape, vanishing as $\sigma_\varepsilon \rightarrow 0$ and $\sigma_\varepsilon \rightarrow \infty$. Figure 10 confirms this prediction in a small economy ($N = 3$, $F = 4$) where exact Jacobian computation is feasible. The peak $\gamma_{\max} = 0.951 < 1$ confirms global BNE uniqueness for all noise scales tested.

The bathtub shape is preserved across economies of different sizes (Figure 11). Two opposing forces determine $\gamma_{\max}$: more firms means smaller market shares and weaker marginal-value sensitivity ($\gamma_{\max}$ decreases with F), while more sites create additional near-marginal activation decisions ($\gamma_{\max}$ increases with N). The contraction condition is most easily satisfied in the empirically relevant regime of many firms with moderate market shares.

F.3 Uniqueness Boundary: AB versus CES

To identify where multiplicity emerges and how AB markups shift the boundary relative to CES, we conduct a joint sweep over TFP dispersion $\delta = a_{\max}/a_{\min}$ and fixed-cost scale α in the synthetic economy.

Figure 12 maps the uniqueness region. The equilibrium is unique across most of the parameter space; multiplicity emerges only at extreme TFP dispersion ($\delta \geq 7$) combined

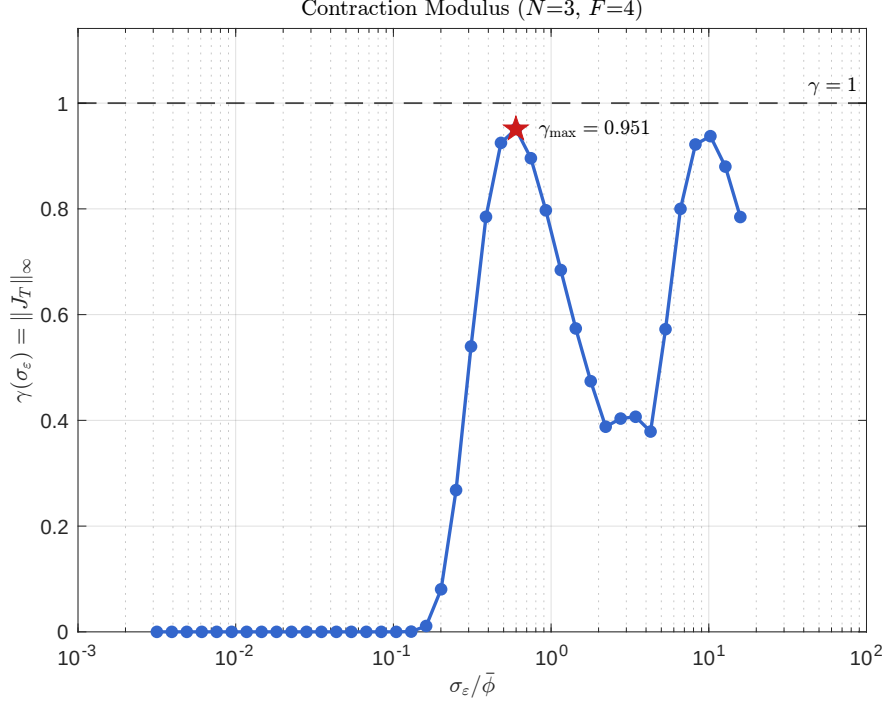


Figure 10: Contraction modulus $\gamma(\sigma_\varepsilon)$ for a small economy ($N = 3, F = 4$). The bathtub shape confirms Proposition 15(iv). Peak $\gamma_{\max} = 0.951$ (marked) is below the $\gamma = 1$ threshold.

with high fixed costs ($\alpha \geq 1.5$), where a single marginal site produces two equilibria.

Figure 13 compares the AB and CES uniqueness regions. On the 11×11 grid, AB markups *expand* the uniqueness region: 23 cells have a unique equilibrium under AB but multiple equilibria under CES, whereas only 1 cell shows the reverse. The mechanism is that AB markup compression dampens the dominant firm's effective demand response, keeping marginal sites' profit gaps bounded away from zero.

F.4 Broader Parameter Sensitivity

Table 15 extends the analysis to a $3 \times 3 \times 4$ grid over (σ, η, θ) . In 18 of 28 configurations the analytic supermodularity condition fails ($(\sigma - 1)/\theta \leq 1$). Nevertheless, the Jia–Tarski gap equals zero and the equilibrium is unique at all 28 configurations. The spectral radius $\rho(\Lambda)$ reaches 0.225 at $(\sigma, \eta) = (8, 1.5)$ but remains well below one throughout.

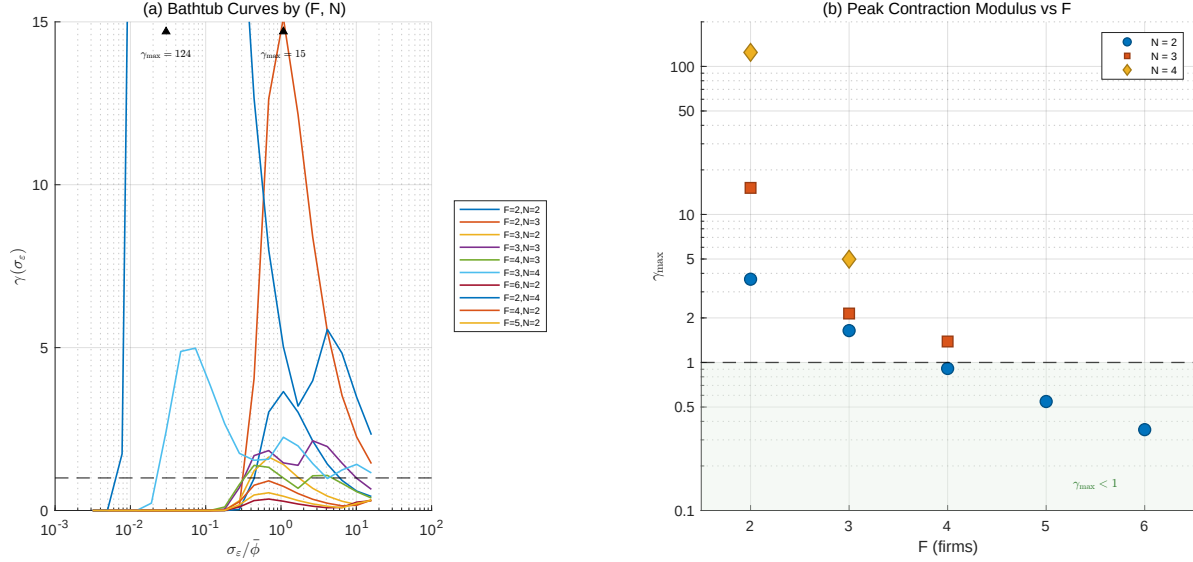


Figure 11: Scaling of the contraction modulus with (F, N) . (a) Bathtub curves for economies of different sizes. (b) Peak $\gamma_{\max}$ vs. F .

F.5 Welfare Decomposition: Variety versus Markup Channel

The AB and CES pricing rules have offsetting welfare effects. Under AB, the 16% activation expansion (37 vs. 32 active sites in the synthetic baseline) generates additional product variety, lowering the sectoral price index P_n . But the higher variable markups ($\bar{\mu}^{AB} = 1.36$ vs. $\mu^{CES} = 1.25$) raise prices. Computing the CES real-income index $W_n = R_n/P_n$ at each market (holding R_n fixed at its initial value), AB welfare is on average 11% *lower* than CES welfare in the synthetic economy. The markup channel dominates the variety channel at all 20 markets, with the welfare gap largest in advanced markets (where concentration is highest) and smallest in frontier markets.

This decomposition underscores that the activation decision—the dimension on which this article’s uniqueness result operates—has quantitatively significant welfare consequences that constant-markup models misattribute to other channels. However, the welfare comparison is partial equilibrium: R_n is held fixed, and general-equilibrium feedbacks through wages or demand could alter the ranking. The result should be interpreted as a diagnostic, not as a normative recommendation.

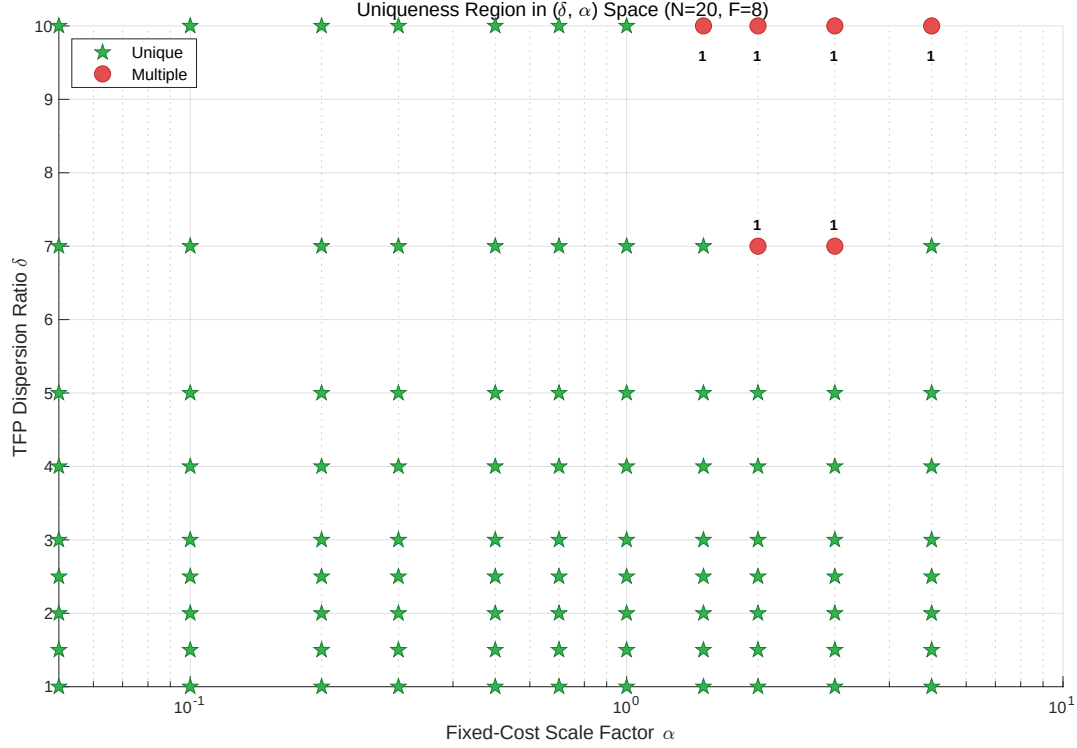


Figure 12: Uniqueness region in (δ, α) space ($N = 20, F = 8$). Green: unique equilibrium; red: multiple equilibria. Multiplicity requires extreme TFP dispersion and high fixed costs.

F.6 Sufficient Conditions: Infinity-Norm versus Spectral Radius

At the synthetic baseline, the maximum feedback coefficient $\max \kappa_{nf} = 0.193$ exceeds the infinity-norm threshold $1/(F - 1) = 0.143$ from Proposition 11, because the dominant firms' high market shares (up to $s = 0.39$, above the per-market threshold $\bar{s} = 0.339$ from Corollary 19) generate large pricing feedback. However, the tighter spectral condition $\rho(\Lambda) = 0.096 < 1$ confirms pricing uniqueness. The same pattern holds in the cloud application ($\rho(\Lambda) = 0.103$): the infinity-norm condition is conservative, while the spectral condition resolves the intensive margin comfortably.

This gap between the two sufficient conditions—the infinity norm violated but the spectral radius well below one—is a generic feature of economies with firm-level TFP heterogeneity. Heterogeneous TFP concentrates market shares in a few dominant firms, inflating $\max \kappa$ and violating the per-market condition, but the spectral radius accounts for the fact that high- κ pairs are few and their feedback does not compound across the full matrix.

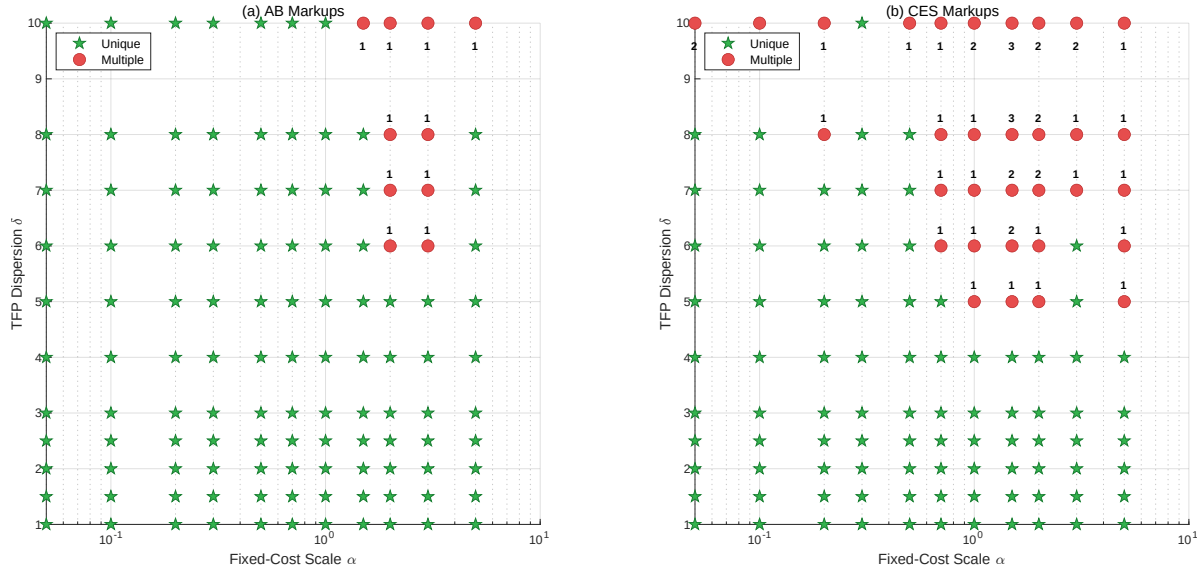


Figure 13: Uniqueness region: AB (left) vs. CES (right). AB markups expand the uniqueness region—23 of 121 cells switch from multiple to unique equilibria.

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Table 15: Uniqueness Diagnostics across (σ, η, θ) : Synthetic Economy

σ	θ	$\eta = 1.5$				$\eta = 2$				$\eta = 3$			
		$\frac{\sigma-1}{\theta}$	ρ	JT	sites	$\frac{\sigma-1}{\theta}$	ρ	JT	sites	$\frac{\sigma-1}{\theta}$	ρ	JT	sites
3	3	0.67	.047	0	39	0.67	.025	0	38				
3	5	0.40	.047	0	19	0.40	.026	0	18				
3	8	0.25	.048	0	16	0.25	.026	0	16				
5	3	1.33	.125	0	38	1.33	.096	0	37	1.33	.053	0	36
5	5	0.80	.123	0	17	0.80	.095	0	16	0.80	.052	0	16
5	8	0.50	.123	0	15	0.50	.095	0	15	0.50	.052	0	15
8	3	2.33	.225	0	39	2.33	.208	0	36	2.33	.160	0	34
8	5	1.40	.222	0	18	1.40	.204	0	18	1.40	.157	0	18
8	8	0.88	.222	0	15	0.88	.204	0	14	0.88	.157	0	13

Notes: Synthetic economy ($N = 20, F = 8$). $\sigma > \eta$ required; $(\sigma, \eta) = (3, 3)$ excluded. JT: Jia–Tarski gap (0 at all configurations). All 28 configurations yield a unique equilibrium.

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